

Volatility Forecasting Report

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Executive Summary

This report presents a comprehensive analysis of volatility forecasting using Heterogeneous Autoregressive (HAR) and HAR with exogenous variables (HAR-X) models applied to Treasury Bond ETF (TLT) data.

Key Objectives: - Evaluate multiple volatility estimators (Squared Return, Parkinson, Garman-Klass, Rogers-Satchell) - Compare HAR and HAR-X model performance across different rolling window sizes - Implement ensemble forecasting using inverse QLIKE weighting - Validate models using out-of-sample testing and statistical comparisons

Main Findings: - Ensemble models outperform individual estimators across all metrics - HAR-X with window=756 provides most stable and consistent predictions - Exogenous variables offer marginal but meaningful improvement in forecast calibration - No statistically significant difference between windows 504 and 756 (DM test) - Both models demonstrate strong forecasting ability with well-behaved residuals

Data Description

Dataset: iShares 20+ Year Treasury Bond ETF (TLT)

Period: 2003-01-01 to 2024-12-30

Frequency: Daily

Total Observations: 5536

Price Data Components: - Open, High, Low, Close prices - Trading volume
- Adjusted close prices

Target Variable: - Realized Volatility (RV): Annualized variance computed from log returns - Log-transformed for modeling to ensure stationarity

Train/Test Split: - Training Set: 2003-01-06 to 2022-12-30 (data before 2023-01-01) — 5033 observations - Test Set: 2023-01-03 to 2024-12-30 (data on/after 2023-01-01) — 501 observations

Methodology

Model Framework

1. Volatility Estimation

Four volatility estimators are computed from OHLC data: - **Squared Return (RV)**: $\sigma^2 = 252 \times (\log(C/O))^2$ - **Parkinson**: $\sigma^2 = 252 \times (1/(4\ln 2)) \times (\log(H/L))^2$ - **Garman-Klass**: $\sigma^2 = 252 \times [0.5(\log(H/L))^2 - (2\ln 2 - 1)(\log(C/O))^2]$ - **Rogers-Satchell**: $\sigma^2 = 252 \times [\log(H/O)\log(H/C) + \log(L/O)\log(L/C)]$

All estimators are log-transformed for modeling.

2. HAR Model

The HAR model captures heterogeneous volatility components:

$$\log(RV) = \omega + \alpha_1 \cdot RV_{\text{Daily}} + \alpha_2 \cdot RV_{\text{Weekly}} + \alpha_3 \cdot RV_{\text{Monthly}}$$

Where: - RV_{Daily} : Daily component (lag 1) - RV_{Weekly} : Weekly component (5-day average) - RV_{Monthly} : Monthly component (22-day average)

3. HAR-X Model

Extends HAR by adding exogenous variables:

$$\log(RV) = \omega + \alpha_1 \cdot RV_{\text{Daily}} + \alpha_2 \cdot RV_{\text{Weekly}} + \alpha_3 \cdot RV_{\text{Monthly}} + \sum \beta_i \cdot X_i$$

Exogenous variables (X_i): - UST10Y (10-Year Treasury Yield) - HYOAS (High Yield Spread) - TermSpread (10Y-2Y) - VIX (Volatility Index) - Breakeven10Y (Inflation expectations)

4. Rolling Window Estimation

Models estimated using rolling windows: 252, 504, 756, 1008, 1260 days

5. Ensemble Forecasting

Predictions combined using inverse QLIKE weighting:

$$w_i = (1/QLIKE_i) / \sum (1/QLIKE_i)$$

Final forecast: $\hat{y} = \sum w_i \times \hat{y}_i$

6. Evaluation Metrics

- **QLIKE**: $\log(\hat{\sigma}^2) + \sigma^2/\hat{\sigma}^2$ (forecast calibration)
- **MSPE**: $((\hat{\sigma}^2 - \sigma^2)/\sigma^2)^2$ (percentage error)
- **RMSE**: $\sqrt{E[(\hat{\sigma}^2 - \sigma^2)^2]}$ (absolute error)
- **Diebold-Mariano Test**: Statistical comparison of forecast accuracy
- **Ljung-Box Test**: Residual autocorrelation check

Volatility Estimators Analysis

Pre-Model Diagnostics

The following table presents the stationarity and autocorrelation diagnostics for each volatility estimator. We use the Augmented Dickey-Fuller (ADF) test and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test to assess stationarity, and the Ljung-Box test to check for serial correlation in the residuals.

Table 1: Diagnostic Tests for Volatility Estimators (Part 1/3)

Estimator	ADF stat	ADF p	ADF pass (p >)	KPSS stat
square_est_log	-9.175	0	True	1.852
parkinson_est_log	-5.068	0	True	0.876
gk_est_log	-4.993	0	True	0.927
rs_est_log	-5.862	0	True	1.251

Table 1: Diagnostic Tests for Volatility Estimators (Part 2/3)

Estimator	KPSS p	KPSS pass (p >)	LB p @10	LB p @20
square_est_log	0.01	False	0	0
parkinson_est_log	0.01	False	0	0
gk_est_log	0.01	False	0	0
rs_est_log	0.01	False	0	0

Table 1: Diagnostic Tests for Volatility Estimators (Part 3/3)

Estimator	White noise (LB)	Stationary (ADF KPSS)
square_est_log	0	0
parkinson_est_log	0	0
gk_est_log	0	0
rs_est_log	0	0

ACF and PACF Analysis

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots reveal: - **Slow decay in ACF**: Indicates long memory in volatility, consistent with volatility clustering - **Significant PACF spikes**: Suggests short-term AR effects up to 5-15 lags - **HAR model justification**: These patterns support using daily (1), weekly (5), and monthly (22) lags

ACF and PACF for All Volatility Estimators

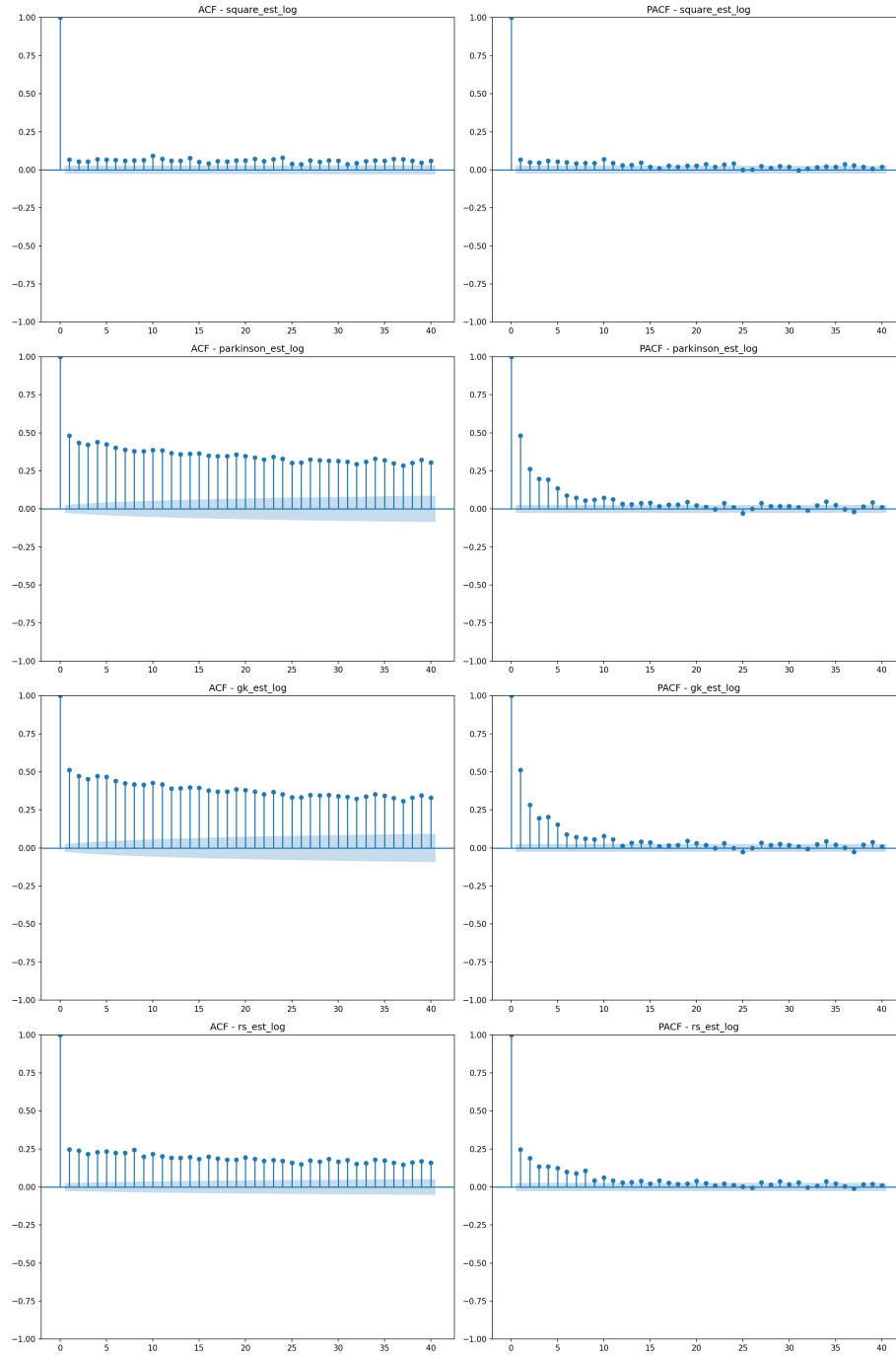


Figure 1: ACF and PACF for All Volatility Estimators

HAR Model Results

Performance Metrics Across Windows

The HAR model was evaluated across multiple rolling window sizes: 252, 504, 756, 1008, and 1260 days. Below are the comprehensive performance metrics for each estimator and window size.

Table 2: HAR Model Performance Summary (QLIKE and MSPE)

	QLIKE_mean	QLIKE_std	MSPE_mean	MSPE_std
(252, 'square_est_log')	-0.928	10.107	1.69095e+17	4.45238e+18
(252, 'parkinson_est_log')	-1.285	7.37	1.83905e+17	4.87268e+18
(252, 'gk_est_log')	-1.282	7.386	1.65485e+17	3.94516e+18
(252, 'rs_est_log')	0.853	147.216	1.5211e+17	3.2143e+18
(504, 'square_est_log')	-0.957	9.524	1.32442e+17	3.20954e+18
(504, 'parkinson_est_log')	-1.217	7.512	1.18736e+17	2.3922e+18
(504, 'gk_est_log')	-1.231	7.317	1.15287e+17	2.21374e+18
(504, 'rs_est_log')	1.047	108.794	1.07971e+17	1.95503e+18
(756, 'square_est_log')	-0.787	10.3	1.1346e+17	2.36631e+18
(756, 'parkinson_est_log')	-1.159	7.449	1.04275e+17	1.99972e+18
(756, 'gk_est_log')	-1.166	7.278	1.02424e+17	1.93991e+18
(756, 'rs_est_log')	2.928	236.818	9.43387e+16	1.68985e+18
(1008, 'square_est_log')	-0.698	10.136	1.32287e+17	2.43004e+18
(1008, 'parkinson_est_log')	-1.105	7.329	1.3284e+17	2.7186e+18
(1008, 'gk_est_log')	-1.113	7.179	1.31281e+17	2.64045e+18
(1008, 'rs_est_log')	-0.774	13.642	1.13775e+17	2.06597e+18
(1260, 'square_est_log')	-0.723	9.852	1.44035e+17	2.49937e+18
(1260, 'parkinson_est_log')	-1.096	7.281	1.49414e+17	2.94141e+18
(1260, 'gk_est_log')	-1.11	7.127	1.48414e+17	2.86121e+18
(1260, 'rs_est_log')	-0.773	13.219	1.28055e+17	2.18766e+18

Table 3: Ljung-Box Test Results for HAR Model Residuals

	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(252, 'lb_stat_10')	18.37382767711334414891951277670467781159382677672771754002	18.37382767711334414891951277670467781159382677672771754002	18.37382767711334414891951277670467781159382677672771754002	18.37382767711334414891951277670467781159382677672771754002
(252, 'lb_p_10')	0.0489757236055808321100003595576583321863263210711560723	0.0489757236055808321100003595576583321863263210711560723	0.0489757236055808321100003595576583321863263210711560723	0.0489757236055808321100003595576583321863263210711560723

	square_est_log	parkinson_est_log	glg_est_log	rs_est_log
(252,	24.468095890858721	7795428481119131	151366775960128	5040537111747
‘lb_stat_20’)				
(252, ‘lb_p_20’)	0.2225424741688588	6641187864006	66202371230378	7907806225183
(252,	False	True	True	True
‘white_noise_flag’)				
(252,	(10, 20)	(10, 20)	(10, 20)	(10, 20)
‘lb_lags_used’)				
(252, ‘n_obs’)	4760	4760	4760	4760
(252, ‘name’)	square_est_log	parkinson_est_log	glg_est_log	rs_est_log
(504,	13.724224989864586	95637640700247	96875999508185	3832204144323
‘lb_stat_10’)				
(504, ‘lb_p_10’)	0.1859462301751176	43593997110374	243936115805746	0294588185283
(504,	20.144932560075124	6509102836618001	8911551906592	0176449150766
‘lb_stat_20’)				
(504, ‘lb_p_20’)	0.448896933955188	7781370737709876	57116485503571	8815282973439
(504,	True	True	True	True
‘white_noise_flag’)				
(504,	(10, 20)	(10, 20)	(10, 20)	(10, 20)
‘lb_lags_used’)				
(504, ‘n_obs’)	4508	4508	4508	4508
(504, ‘name’)	square_est_log	parkinson_est_log	glg_est_log	rs_est_log
(756,	17.50738499144659	675510755236970	447387608868351	79507833852
‘lb_stat_10’)				
(756, ‘lb_p_10’)	0.063864079281406	029834127270697	2972644670332	2141729716968
(756,	21.941585926708514	98838109551650	3128511949563	47405173655635
‘lb_stat_20’)				
(756, ‘lb_p_20’)	0.343688748375259	16475316099829	842078312671717	4383814594563
(756,	True	True	True	True
‘white_noise_flag’)				
(756,	(10, 20)	(10, 20)	(10, 20)	(10, 20)
‘lb_lags_used’)				
(756, ‘n_obs’)	4256	4256	4256	4256
(756, ‘name’)	square_est_log	parkinson_est_log	glg_est_log	rs_est_log
(1008,	20.83105657308165	99059938928997	698596799363	541889982428769
‘lb_stat_10’)				
(1008,	0.022302911899427	5034431608807025	1470891802879	3114186075355
‘lb_p_10’)				
(1008,	25.038978922032913	90577698162137	9239044017729	067118807268
‘lb_stat_20’)				
(1008,	0.199943937518328	690582192290165	3797681263178	0835436406335
‘lb_p_20’)				
(1008,	False	True	True	True
‘white_noise_flag’)				

	square_est_log	parkinson_est_log	gl_est_log	rs_est_log
(1008, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(1008, 'n_obs')	4004	4004	4004	4004
(1008, 'name')	square_est_log	parkinson_est_log	gl_est_log	rs_est_log
(1260, 'lb_stat_10')	17.460719088070742	10.23558442283041381791709	10.40533257560807	
(1260, 'lb_p_10')	0.06477105475142826	0.32094147568708216887061389	0.177486719972	
(1260, 'lb_stat_20')	22.035485989335092	335945903919261	386706862171206559634457115	
(1260, 'lb_p_20')	0.33858818502575816	4573395016379	8554408096528	4381672944962
(1260, 'white_noise_flag')	True	True	True	True
(1260, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(1260, 'n_obs')	3752	3752	3752	3752
(1260, 'name')	square_est_log	parkinson_est_log	gl_est_log	rs_est_log

HAR Model Predictions vs True RV

The following plots compare the predicted volatility from each estimator against the true realized volatility.

HAR Model: Predictions vs True RV (Window=252)

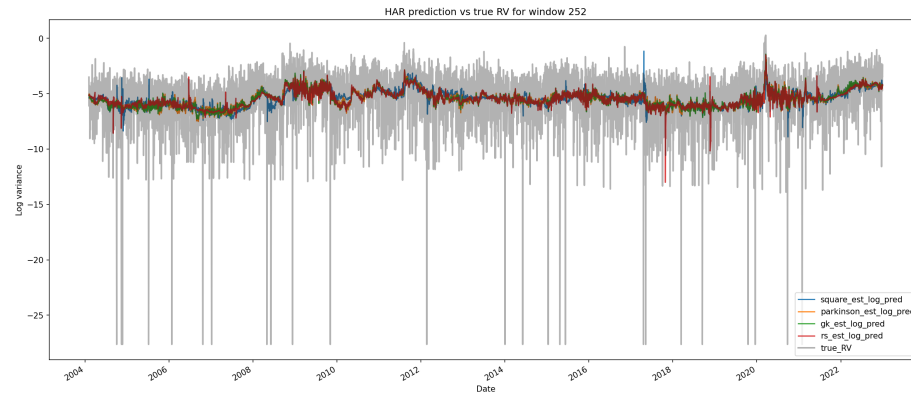


Figure 2: HAR Model: Predictions vs True RV (Window=252)

HAR Model: Predictions vs True RV (Window=504)

HAR Model: Predictions vs True RV (Window=756)

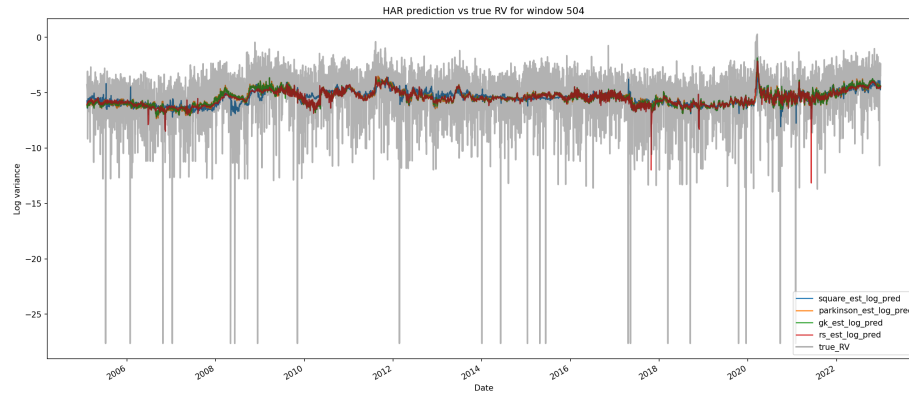


Figure 3: HAR Model: Predictions vs True RV (Window=504)

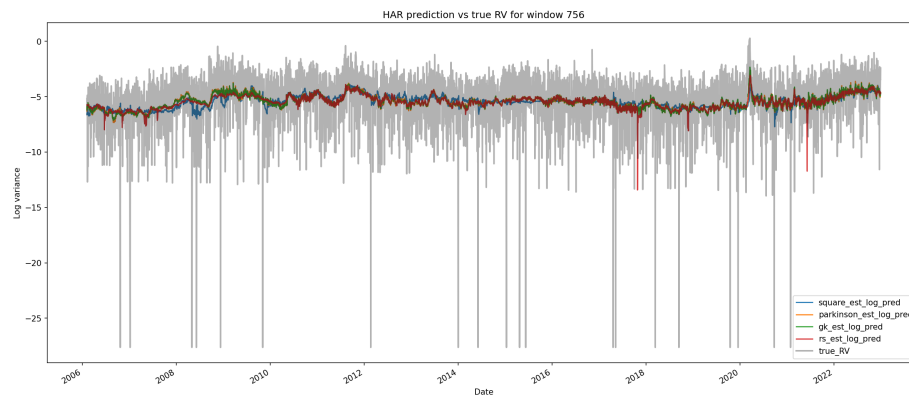


Figure 4: HAR Model: Predictions vs True RV (Window=756)

HAR Model: Predictions vs True RV (Window=1008)

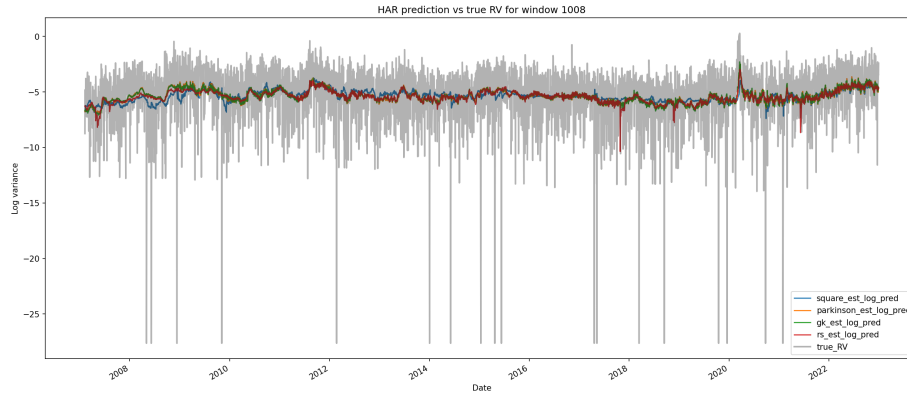


Figure 5: HAR Model: Predictions vs True RV (Window=1008)

HAR Model: Predictions vs True RV (Window=1260)

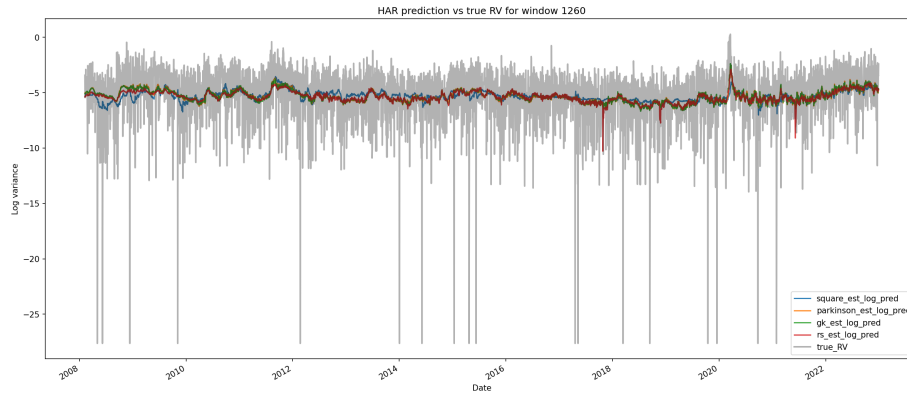


Figure 6: HAR Model: Predictions vs True RV (Window=1260)

Loss Metrics Over Time

QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude.

QLIKE Loss Over Time (Window=252)

QLIKE Loss Over Time (Window=504)

QLIKE Loss Over Time (Window=756)

QLIKE Loss Over Time (Window=1008)

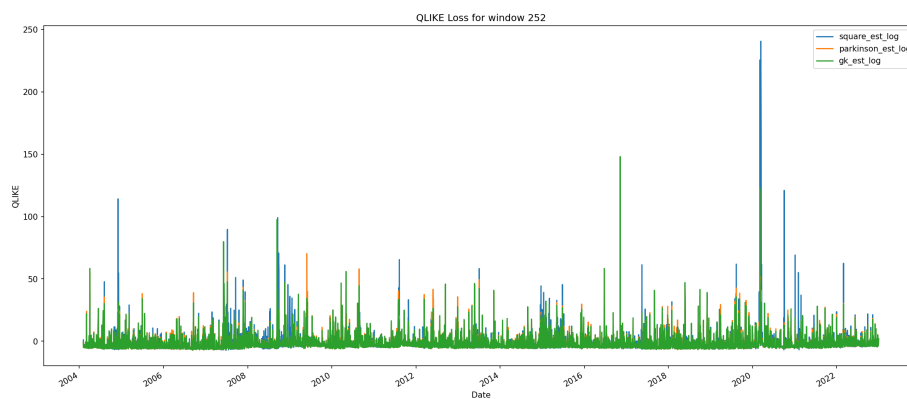


Figure 7: QLIKE Loss Over Time (Window=252)

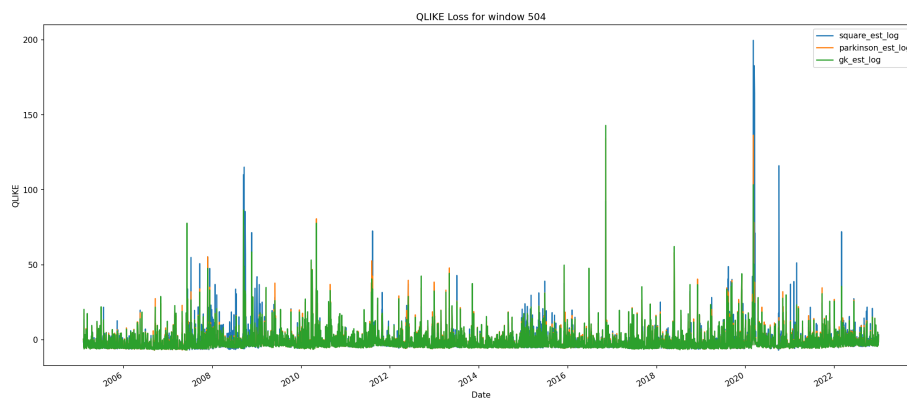


Figure 8: QLIKE Loss Over Time (Window=504)

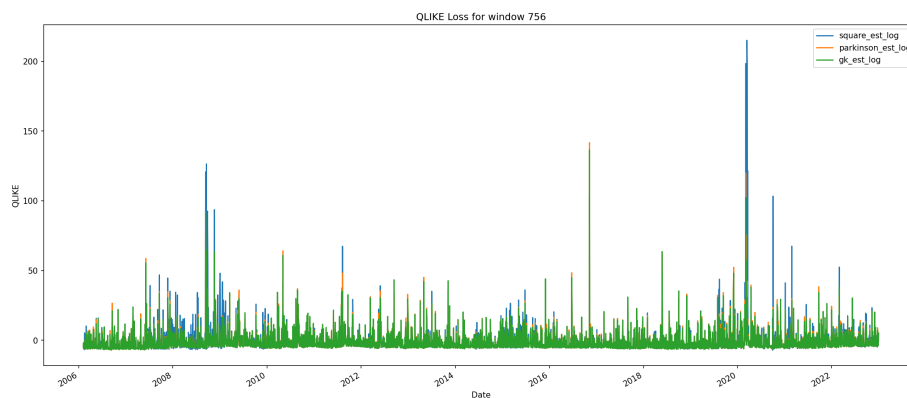


Figure 9: QLIKE Loss Over Time (Window=756)

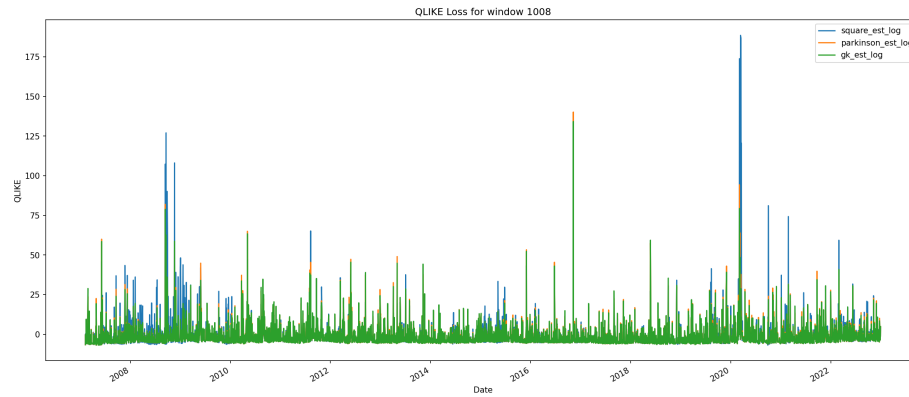


Figure 10: QLIKE Loss Over Time (Window=1008)

QLIKE Loss Over Time (Window=1260)

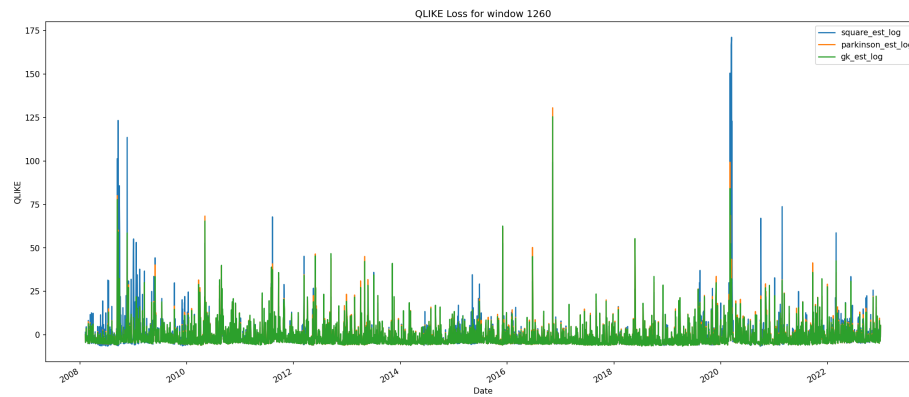


Figure 11: QLIKE Loss Over Time (Window=1260)

MSPE Loss Over Time (Window=252)

MSPE Loss Over Time (Window=504)

MSPE Loss Over Time (Window=756)

MSPE Loss Over Time (Window=1008)

MSPE Loss Over Time (Window=1260)

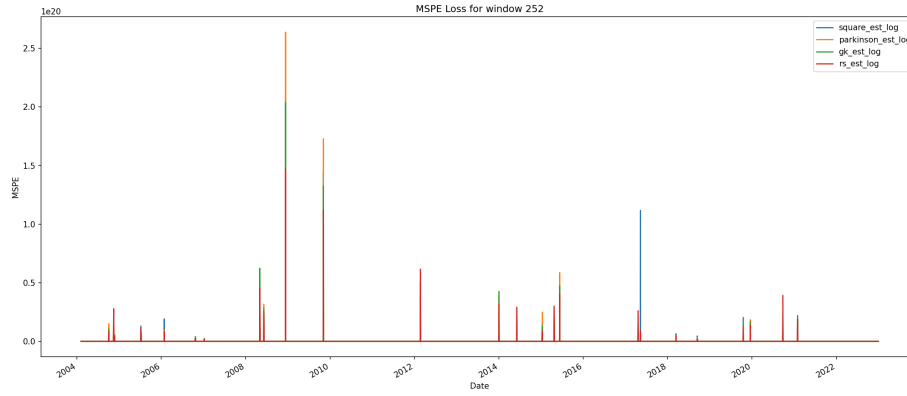


Figure 12: MSPE Loss Over Time (Window=252)

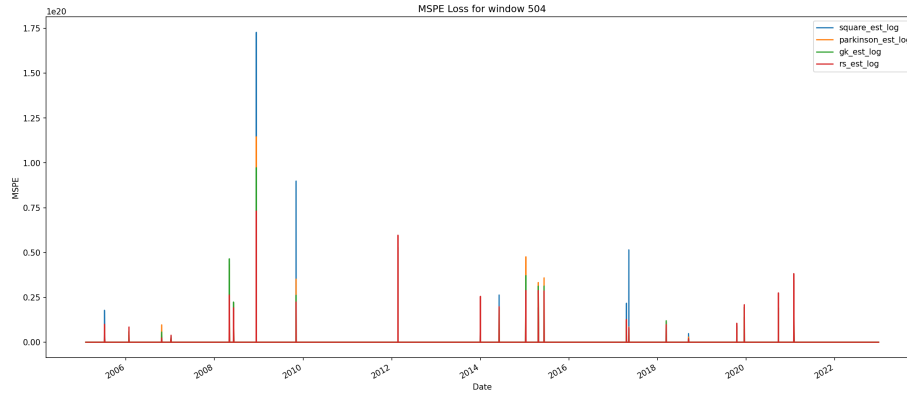


Figure 13: MSPE Loss Over Time (Window=504)

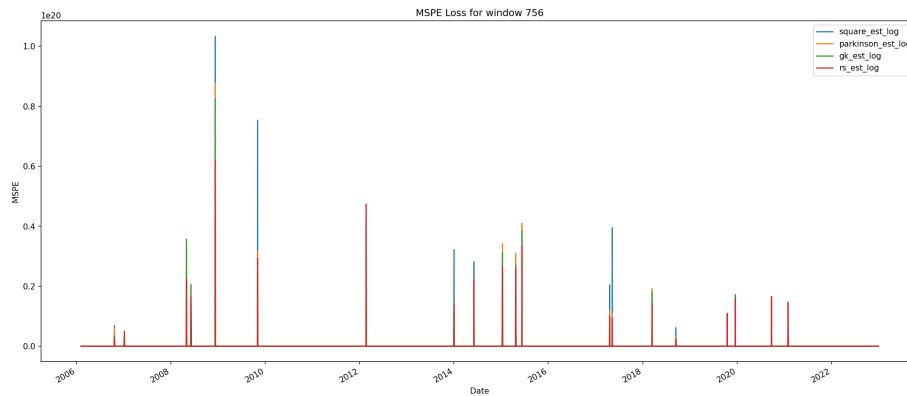


Figure 14: MSPE Loss Over Time (Window=756)

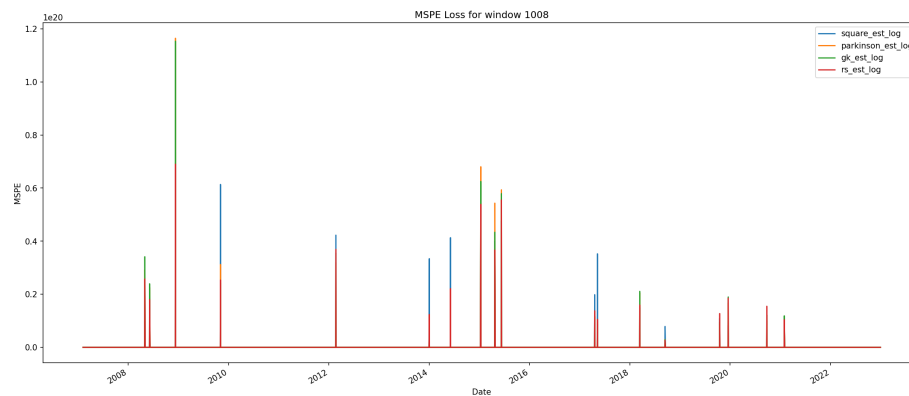


Figure 15: MSPE Loss Over Time (Window=1008)

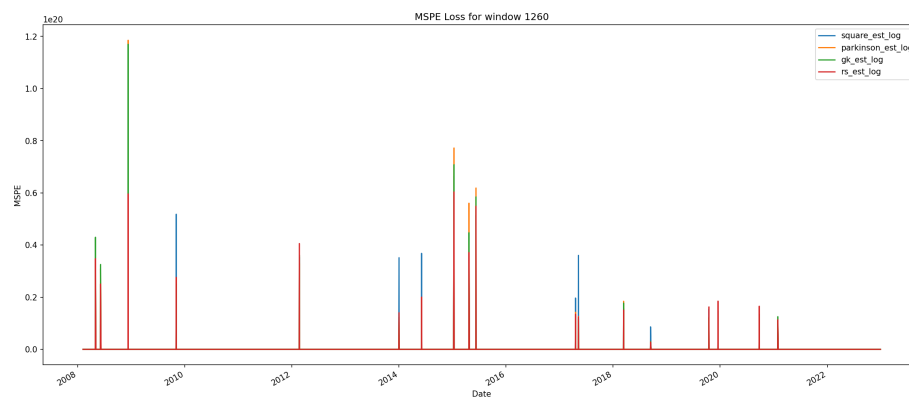


Figure 16: MSPE Loss Over Time (Window=1260)

Ensemble Model Results

Ensemble Weights

The ensemble model combines predictions from multiple estimators using inverse QLIKE weighting. Below are the weights assigned to each estimator for different window sizes.

Table 4: Ensemble Model Weights by Window

Window	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
252	0.333	0.333	0.333	0
504	0.333	0.333	0.333	0
756	0.333	0.333	0.333	0
1008	0.25	0.25	0.25	0.25
1260	0.25	0.25	0.25	0.25

Ensemble Performance Summary

Table 5: Ensemble Model Performance Metrics

	QLIKE_mean	QLIKE_std	MSPE_mean	MSPE_std
(252, 0)	-1.382	7.18	1.68049e+17	4.29666e+18
(504, 0)	-1.314	7.195	1.18338e+17	2.45339e+18
(756, 0)	-1.217	7.342	1.04067e+17	2.00377e+18
(1008, 0)	-1.143	7.218	1.25111e+17	2.37633e+18
(1260, 0)	-1.139	7.18	1.39999e+17	2.5294e+18

Table 6: Ensemble Model Ljung-Box Test Results (Part 1/2)

	lb_stat_10	lb_p_10	lb_stat_20	lb_p_20
(252, 0)	7.565	0.671	12.928	0.88
(252, 1)	7.565	0.671	12.928	0.88
(504, 0)	6.153	0.802	11.937	0.918
(504, 1)	6.153	0.802	11.937	0.918
(756, 0)	7.398	0.687	11.344	0.937
(756, 1)	7.398	0.687	11.344	0.937
(1008, 0)	9.009	0.531	12.804	0.886
(1008, 1)	9.009	0.531	12.804	0.886
(1260, 0)	9.259	0.508	13.255	0.866
(1260, 1)	9.259	0.508	13.255	0.866

Table 6: Ensemble Model Ljung-Box Test Results (Part 2/2)

	white_noise_flag	lb_lags_used	n_obs	name
(252, 0)	True	10	4760	
(252, 1)	True	20	4760	
(504, 0)	True	10	4508	
(504, 1)	True	20	4508	
(756, 0)	True	10	4256	
(756, 1)	True	20	4256	
(1008, 0)	True	10	4004	
(1008, 1)	True	20	4004	
(1260, 0)	True	10	3752	
(1260, 1)	True	20	3752	

Ensemble Predictions and Loss Metrics

Ensemble Model: Predictions vs True RV (Window=252)

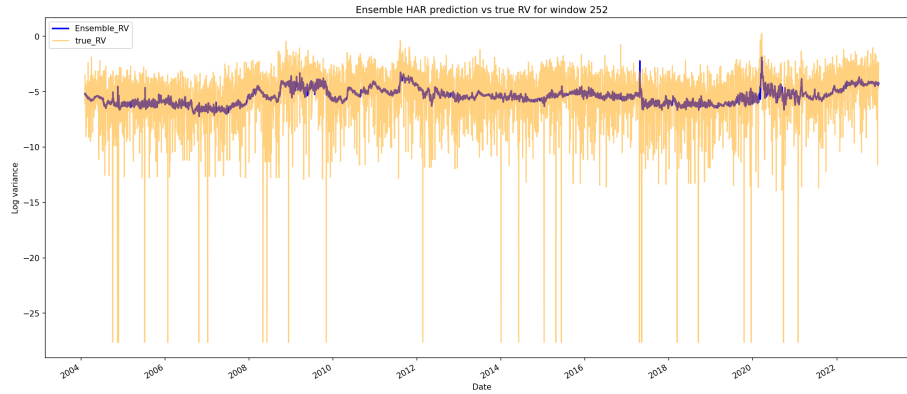


Figure 17: Ensemble Model: Predictions vs True RV (Window=252)

Ensemble Model: Predictions vs True RV (Window=504)

Ensemble Model: Predictions vs True RV (Window=756)

Ensemble Model: Predictions vs True RV (Window=1008)

Ensemble Model: Predictions vs True RV (Window=1260)

Ensemble QLIKE Loss (Window=252)

Ensemble QLIKE Loss (Window=504)

Ensemble QLIKE Loss (Window=756)

Ensemble QLIKE Loss (Window=1008)

Ensemble QLIKE Loss (Window=1260)

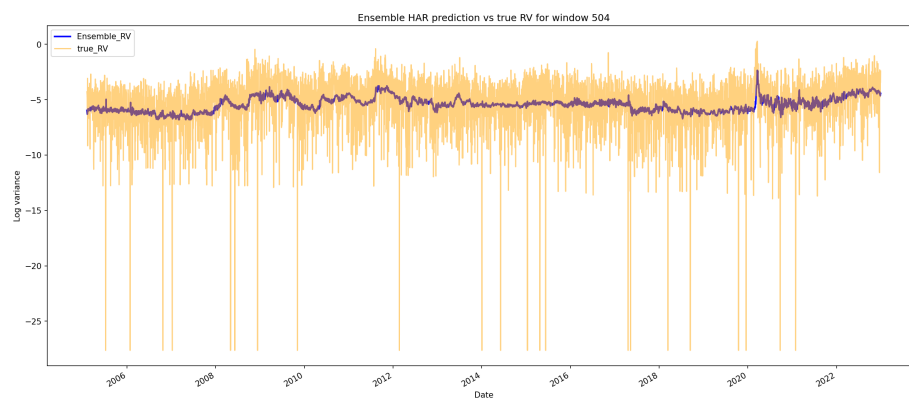


Figure 18: Ensemble Model: Predictions vs True RV (Window=504)

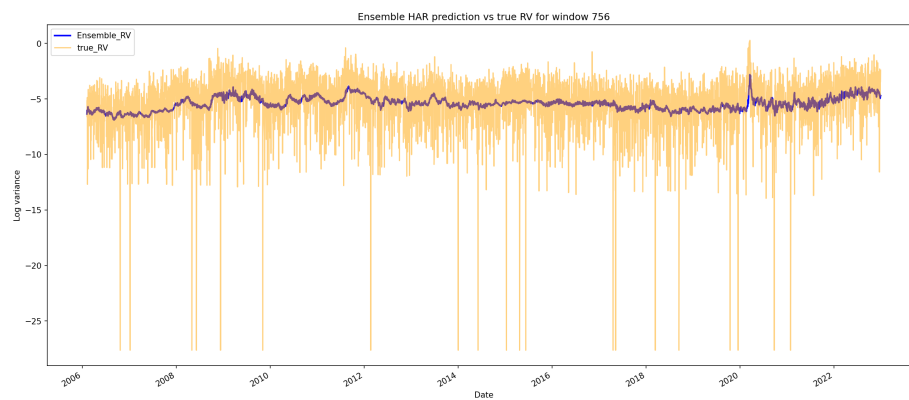


Figure 19: Ensemble Model: Predictions vs True RV (Window=756)

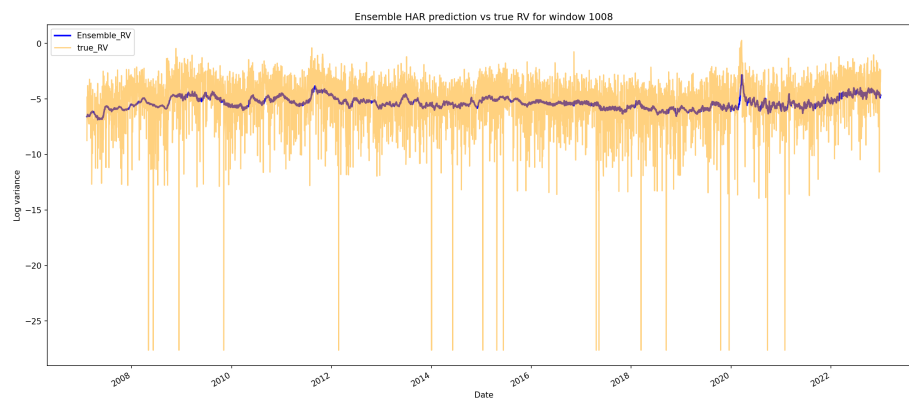


Figure 20: Ensemble Model: Predictions vs True RV (Window=1008)

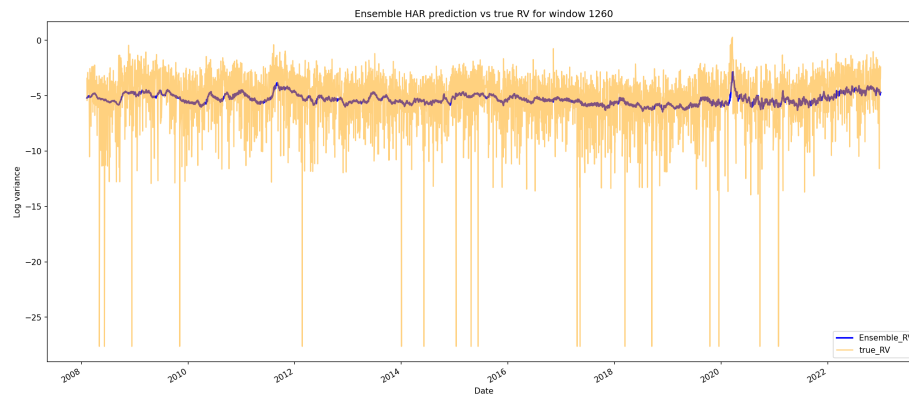


Figure 21: Ensemble Model: Predictions vs True RV (Window=1260)

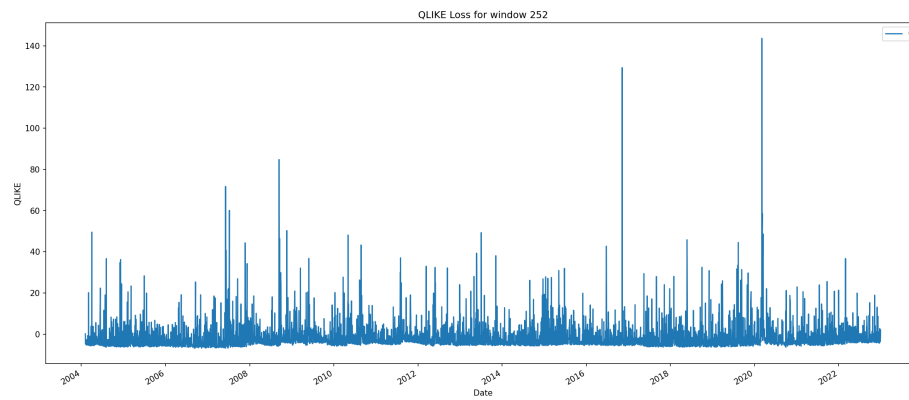


Figure 22: Ensemble QLIKE Loss (Window=252)

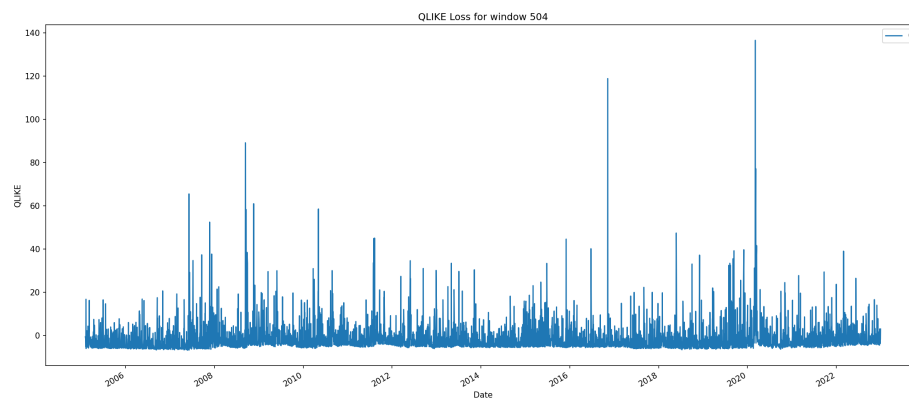


Figure 23: Ensemble QLIKE Loss (Window=504)

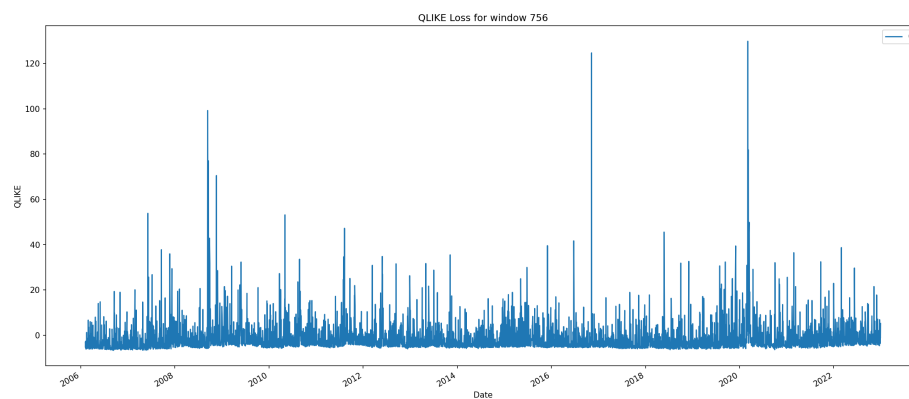


Figure 24: Ensemble QLIKE Loss (Window=756)

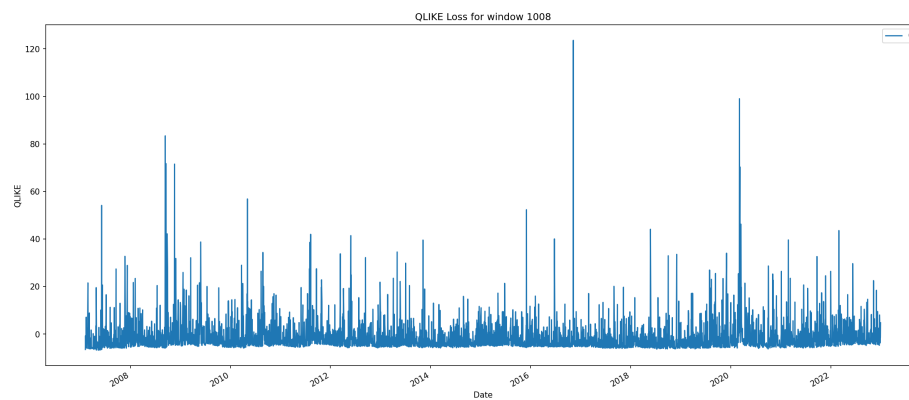


Figure 25: Ensemble QLIKE Loss (Window=1008)

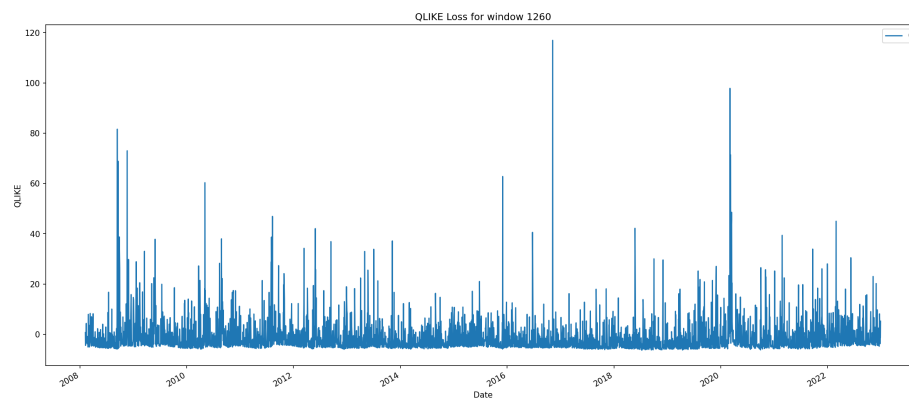


Figure 26: Ensemble QLIKE Loss (Window=1260)

Ensemble MSPE Loss (Window=252)

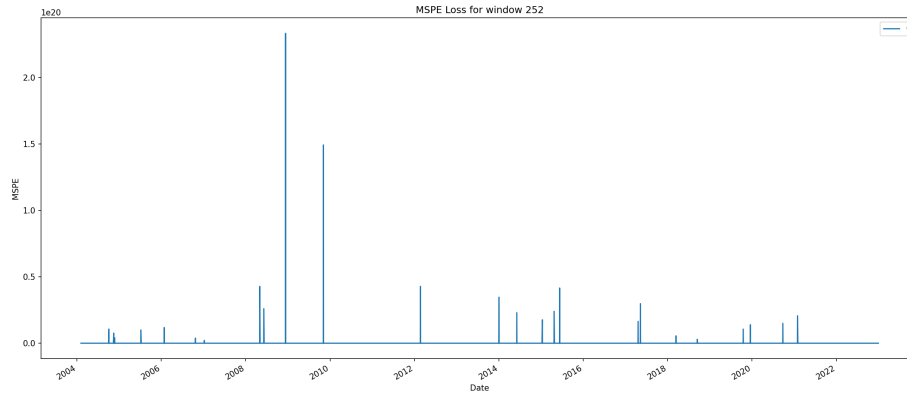


Figure 27: Ensemble MSPE Loss (Window=252)

Ensemble MSPE Loss (Window=504)

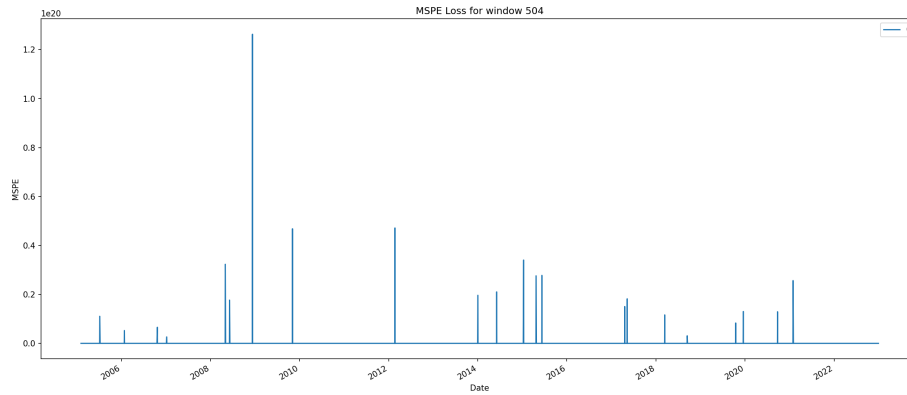


Figure 28: Ensemble MSPE Loss (Window=504)

Ensemble MSPE Loss (Window=756)

Ensemble MSPE Loss (Window=1008)

Ensemble MSPE Loss (Window=1260)

Diebold-Mariano Test

The Diebold-Mariano (DM) test evaluates whether there is a statistically significant difference in predictive accuracy between two models. Here we compare Windows 504 and 756.

DM Test Results: Window 504 vs Window 756

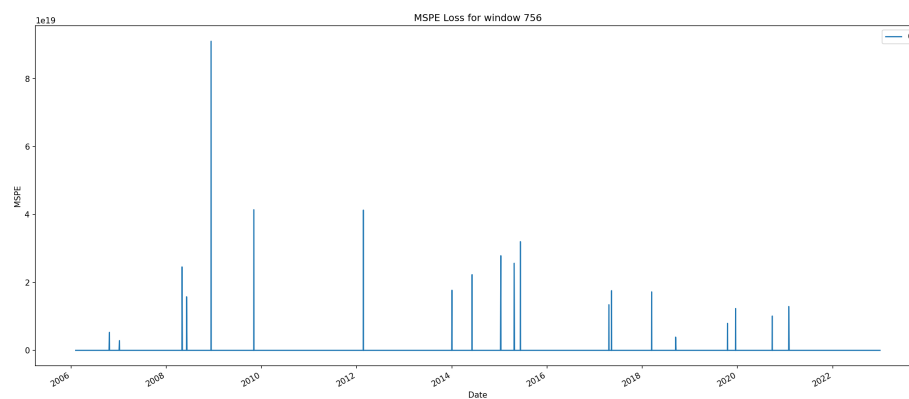


Figure 29: Ensemble MSPE Loss (Window=756)

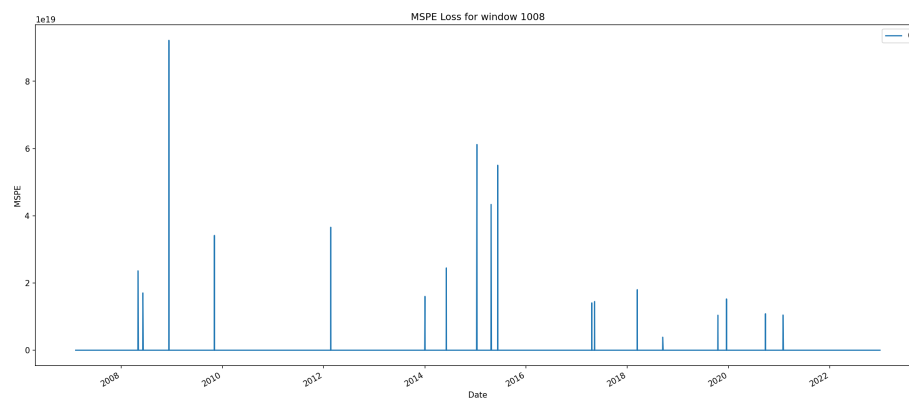


Figure 30: Ensemble MSPE Loss (Window=1008)

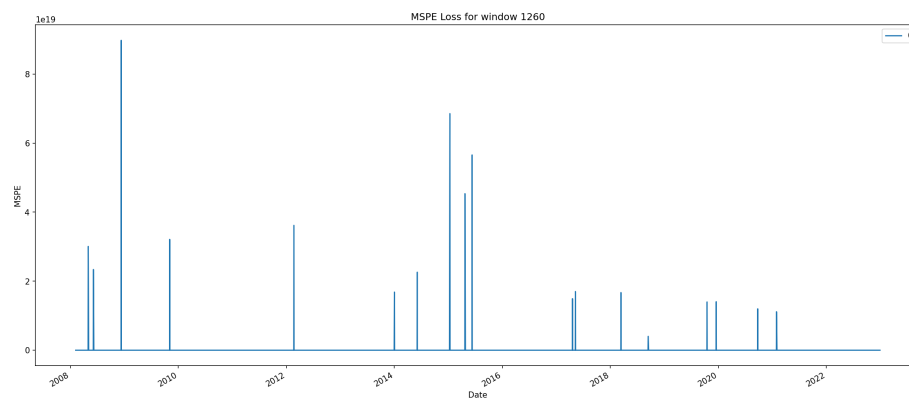


Figure 31: Ensemble MSPE Loss (Window=1260)

Metric	Value
DM Statistic	-0.4968
P-value	0.6194
Better Model	None (No significant difference)
Significant?	0.0000
Alpha	0.0500
Observations	4256.0000

Interpretation: With $p\text{-value} = 0.6194 > 0.05$, we fail to reject the null hypothesis of equal predictive accuracy. This indicates no statistically significant difference between Window 504 and 756. Either window may be used, and selection can be based on secondary metrics or practical considerations.

HAR-X Model Results

Exogenous Variables

The HAR-X model extends the HAR model by incorporating exogenous variables: - **UST10Y**: 10-Year US Treasury Yield - **HYOAS**: High Yield Option-Adjusted Spread - **TermSpread_10Y_2Y**: Term Spread (10Y - 2Y) - **VIX**: CBOE Volatility Index - **Breakeven10Y**: 10-Year Breakeven Inflation Rate

All exogenous variables were standardized using expanding window standardization to prevent look-ahead bias.

Table 7: Diagnostic Tests for Exogenous Variables (After Differencing) (Part 1/3)

Estimator	ADF stat	ADF p	ADF pass (p >)	KPSS stat
UST10Y	-3.431	0.01	True	1.567
HYOAS	-3.298	0.015	True	0.882
TermSpread_10Y_2Y	-17.061	0	True	0.048
VIX	-5.957	0	True	0.463
Breakeven10Y	-4.528	0	True	0.823

Table 7: Diagnostic Tests for Exogenous Variables (After Differencing) (Part 2/3)

Estimator	KPSS p	KPSS pass (p >)	LB p @10	LB p @20
UST10Y	0.01	False	0	0
HYOAS	0.01	False	0	0
TermSpread_10Y_2Y	0.1	True	0	0
VIX	0.05	False	0	0

Estimator	KPSS p	KPSS pass (p>)	LB p @10	LB p @20
Breakeven10Y	0.01	False	0	0

Table 7: Diagnostic Tests for Exogenous Variables (After Differencing) (Part 3/3)

Estimator	White noise (LB)	Stationary (ADF KPSS)
UST10Y	0	0
HYOAS	0	0
TermSpread_10Y_2Y	0	1
VIX	0	0
Breakeven10Y	0	0

HARX Model Performance

The HAR-X model performance across different rolling window sizes is presented below.

Table 8: HAR-X Model Performance Summary

	QLIKE_mean	QLIKE_std	MSPE_mean	MSPE_std
(252, 'square_est_log')	-1.038	9.387	3.19309e+17	8.28011e+18
(252, 'parkinson_est_log')	-1.084	9.472	3.24395e+17	8.91223e+18
(252, 'gk_est_log')	-1.056	9.64	3.07645e+17	8.18992e+18
(252, 'rs_est_log')	-0.842	21.162	2.90226e+17	6.9799e+18
(504, 'square_est_log')	-1.306	7.365	2.30542e+17	5.51343e+18
(504, 'parkinson_est_log')	-1.255	7.54	2.02421e+17	4.52756e+18
(504, 'gk_est_log')	-1.242	7.484	1.96227e+17	4.22809e+18
(504, 'rs_est_log')	-0.788	25.674	2.10912e+17	4.82157e+18
(756, 'square_est_log')	-1.328	6.802	2.53527e+17	7.23164e+18
(756, 'parkinson_est_log')	-1.308	6.915	2.10233e+17	5.68918e+18
(756, 'gk_est_log')	-1.308	6.849	2.10418e+17	5.72007e+18
(756, 'rs_est_log')	0.174	93.545	2.20903e+17	6.08202e+18
(1008, 'square_est_log')	-1.195	6.866	2.24376e+17	5.49743e+18
(1008, 'parkinson_est_log')	-1.237	6.838	2.00567e+17	4.63773e+18
(1008, 'gk_est_log')	-1.246	6.775	2.08867e+17	4.98968e+18
(1008, 'rs_est_log')	-1.131	8.288	2.10246e+17	5.03113e+18

	QLIKE_mean	QLIKE_std	MSPE_mean	MSPE_std
(1260, 'square_est_log')	-1.167	7.056	2.23941e+17	5.08719e+18
(1260, 'parkinson_est_log')	-1.22	6.934	2.01552e+17	4.20515e+18
(1260, 'gk_est_log')	-1.228	6.913	2.13547e+17	4.63074e+18
(1260, 'rs_est_log')	-1.075	9.032	2.02133e+17	4.24679e+18

Table 9: HAR-X Model Ljung-Box Test Results

	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(252, 'lb_stat_10')	36.04321663663161179662381481624255106577030268	36.04321663663161179662381481624255106577030268	36.04321663663161179662381481624255106577030268	36.04321663663161179662381481624255106577030268
(252, 'lb_p_10')	8.274866978845706734841607950274923200938104736	8.274866978845706734841607950274923200938104736	8.274866978845706734841607950274923200938104736	8.274866978845706734841607950274923200938104736
(252, 'lb_stat_20')	46.2526905399011253619704478778960408554321217	46.2526905399011253619704478778960408554321217	46.2526905399011253619704478778960408554321217	46.2526905399011253619704478778960408554321217
(252, 'lb_p_20')	0.00074408169110276335978756028374255109670886	0.00074408169110276335978756028374255109670886	0.00074408169110276335978756028374255109670886	0.00074408169110276335978756028374255109670886
(252, 'white_noise_flag')	False	True	True	True
(252, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(252, 'n_obs')	5257	5257	5257	5257
(252, 'name')	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(504, 'lb_stat_10')	13.193500632597679316677513252706262781032700	13.193500632597679316677513252706262781032700	13.193500632597679316677513252706262781032700	13.193500632597679316677513252706262781032700
(504, 'lb_p_10')	0.213053345088107725089893895082967295589301386	0.213053345088107725089893895082967295589301386	0.213053345088107725089893895082967295589301386	0.213053345088107725089893895082967295589301386
(504, 'lb_stat_20')	21.20671621224003749771598541413152246188857848	21.20671621224003749771598541413152246188857848	21.20671621224003749771598541413152246188857848	21.20671621224003749771598541413152246188857848
(504, 'lb_p_20')	0.38505645675002752295304632095741604081395804	0.38505645675002752295304632095741604081395804	0.38505645675002752295304632095741604081395804	0.38505645675002752295304632095741604081395804
(504, 'white_noise_flag')	True	True	True	True
(504, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(504, 'n_obs')	5005	5005	5005	5005
(504, 'name')	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(756, 'lb_stat_10')	18.0299537401080015896292294385637237162911142	18.0299537401080015896292294385637237162911142	18.0299537401080015896292294385637237162911142	18.0299537401080015896292294385637237162911142
(756, 'lb_p_10')	0.0544604616884073703697268410293646974034900325	0.0544604616884073703697268410293646974034900325	0.0544604616884073703697268410293646974034900325	0.0544604616884073703697268410293646974034900325

	square_est_log	parkinson_est_log	glog_est_log	rs_est_log
(756, 'lb_stat_20')	25.97513271619996	3370916832010652	2951091115613	73154743497398
(756, 'lb_p_20')	0.16663473982500695	5037487990483	2789421972532	55890884649988
(756, 'white_noise_flag')	True	True	True	True
(756, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(756, 'n_obs')	4753	4753	4753	4753
(756, 'name')	square_est_log	parkinson_est_log	glog_est_log	rs_est_log
(1008, 'lb_stat_10')	21.393728624987872	352371087366155	3385037533962	3431046517541
(1008, 'lb_p_10')	0.0185093517009442972	58290864534235	7278665702777	6265186330845
(1008, 'lb_stat_20')	28.21281544140536	602903425181407	11771351529237	642855663289
(1008, 'lb_p_20')	0.104456530376382378	58770519165715	9495107922652	85671790449226
(1008, 'white_noise_flag')	False	True	True	True
(1008, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(1008, 'n_obs')	4501	4501	4501	4501
(1008, 'name')	square_est_log	parkinson_est_log	glog_est_log	rs_est_log
(1260, 'lb_stat_10')	18.82776344201063	399641105273234	1865444520264	8238889267308
(1260, 'lb_p_10')	0.042505677015862524	19979122761258	3121311236778	8026332467069
(1260, 'lb_stat_20')	26.533444096812296	183303005725822	2076361126396	40439979210182
(1260, 'lb_p_20')	0.148907743063203672	33345105746574	28680797182983	9082051312682
(1260, 'white_noise_flag')	False	True	True	True
(1260, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(1260, 'n_obs')	4249	4249	4249	4249
(1260, 'name')	square_est_log	parkinson_est_log	glog_est_log	rs_est_log

HAR-X Model Predictions vs True RV

The following plots compare the predicted volatility from each estimator with exogenous variables against the true realized volatility.

HAR-X Model: Predictions vs True RV (Window=252)

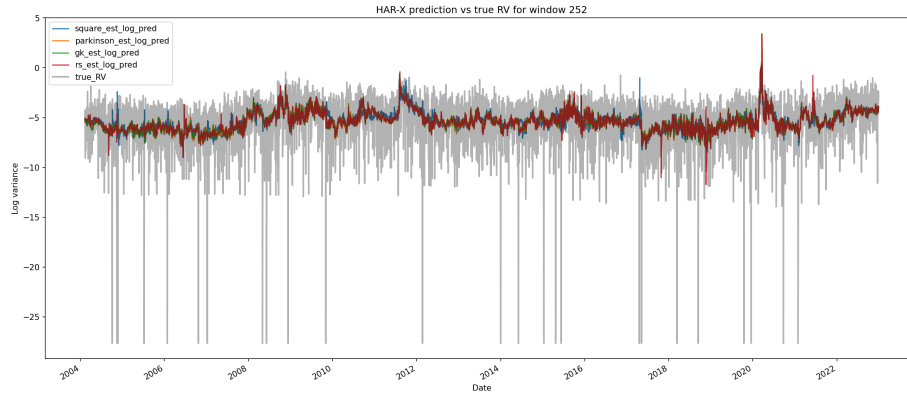


Figure 32: HAR-X Model: Predictions vs True RV (Window=252)

HAR-X Model: Predictions vs True RV (Window=504)

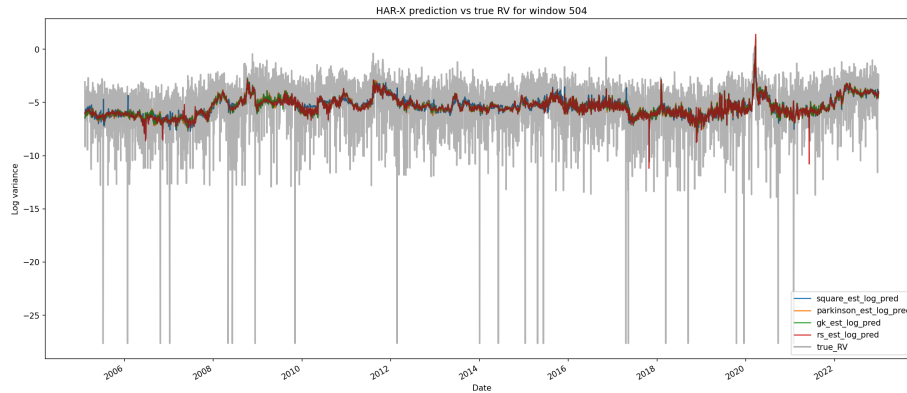


Figure 33: HAR-X Model: Predictions vs True RV (Window=504)

HAR-X Model: Predictions vs True RV (Window=756)

HAR-X Model: Predictions vs True RV (Window=1008)

HAR-X Model: Predictions vs True RV (Window=1260)

Loss Metrics Over Time

QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude for the HAR-X model with exogenous variables.

HAR-X QLIKE Loss Over Time (Window=252)

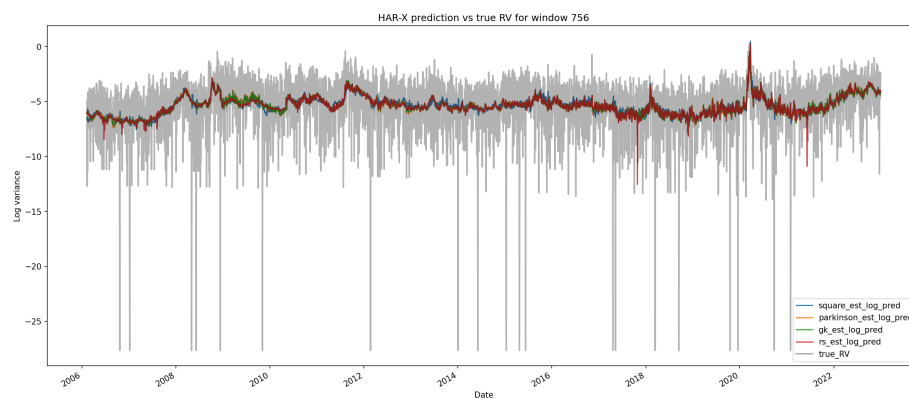


Figure 34: HAR-X Model: Predictions vs True RV (Window=756)

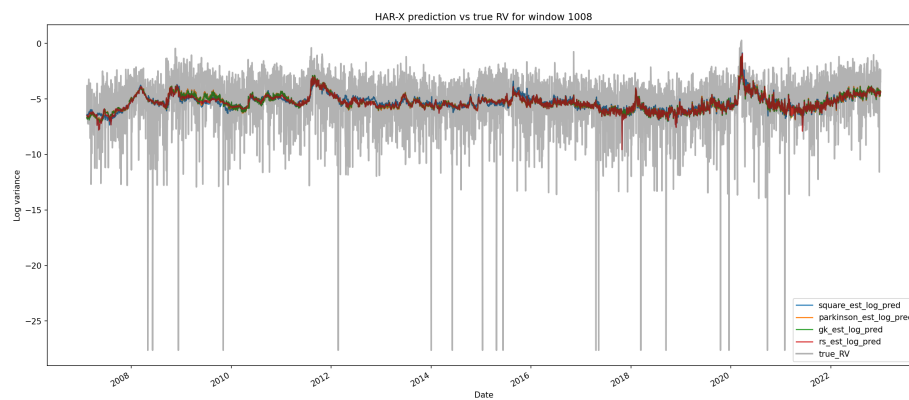


Figure 35: HAR-X Model: Predictions vs True RV (Window=1008)

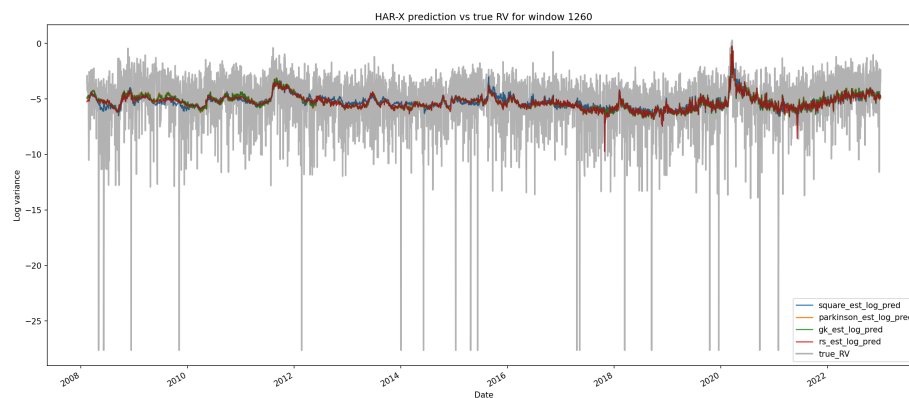


Figure 36: HAR-X Model: Predictions vs True RV (Window=1260)

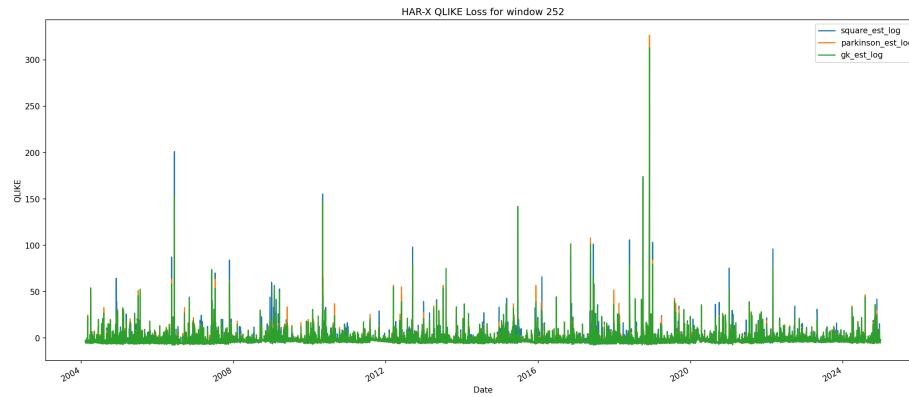


Figure 37: HAR-X QLIKE Loss Over Time (Window=252)

HAR-X QLIKE Loss Over Time (Window=504)

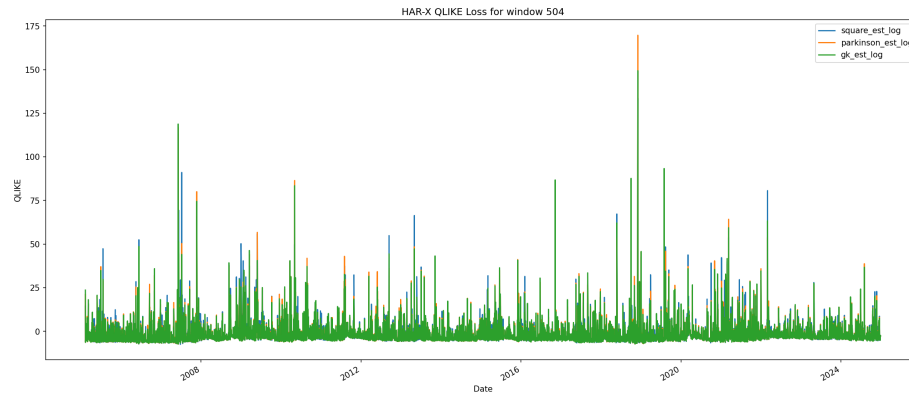


Figure 38: HAR-X QLIKE Loss Over Time (Window=504)

HAR-X QLIKE Loss Over Time (Window=756)

HAR-X QLIKE Loss Over Time (Window=1008)

HAR-X QLIKE Loss Over Time (Window=1260)

HAR-X MSPE Loss Over Time (Window=252)

HAR-X MSPE Loss Over Time (Window=504)

HAR-X MSPE Loss Over Time (Window=756)

HAR-X MSPE Loss Over Time (Window=1008)

HAR-X MSPE Loss Over Time (Window=1260)

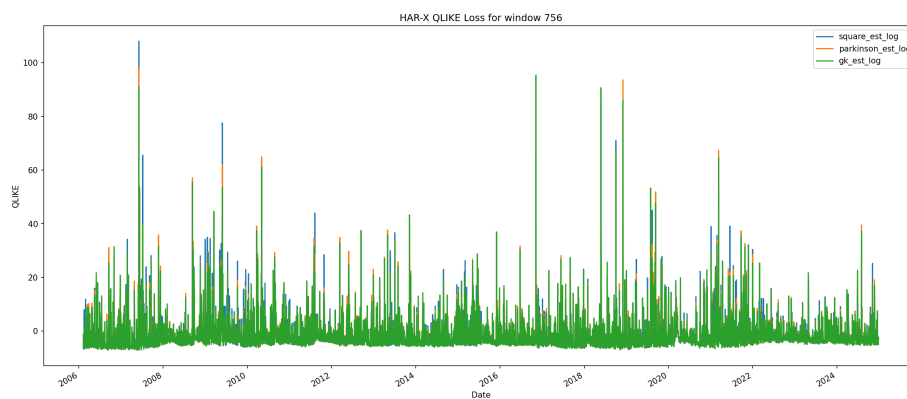


Figure 39: HAR-X QLIKE Loss Over Time (Window=756)

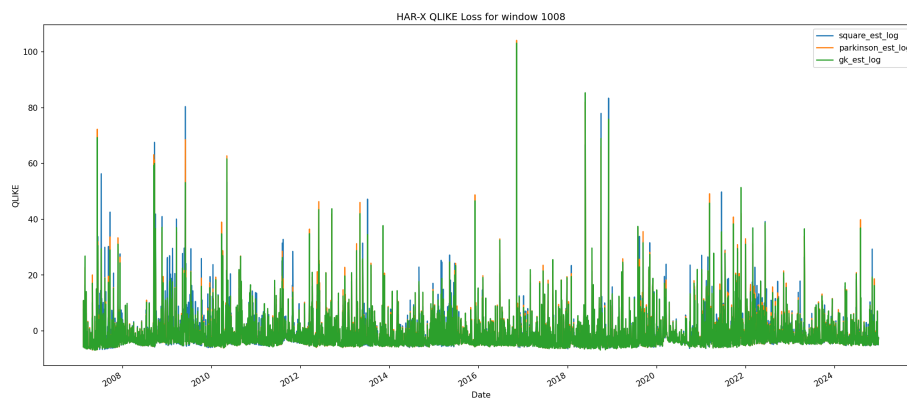


Figure 40: HAR-X QLIKE Loss Over Time (Window=1008)

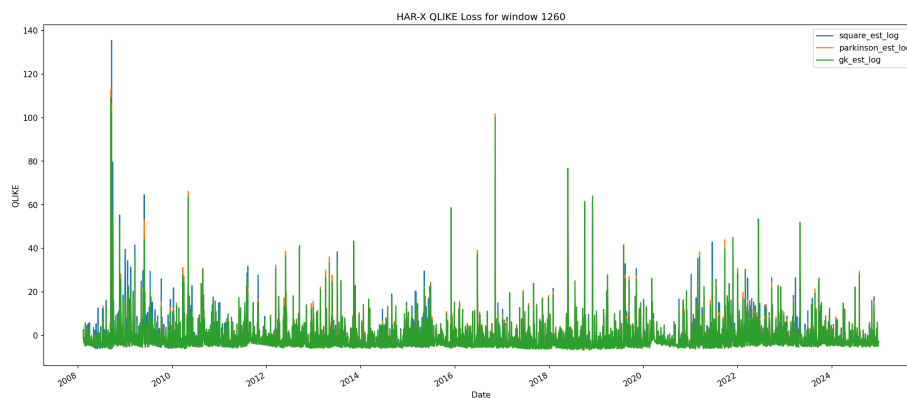


Figure 41: HAR-X QLIKE Loss Over Time (Window=1260)

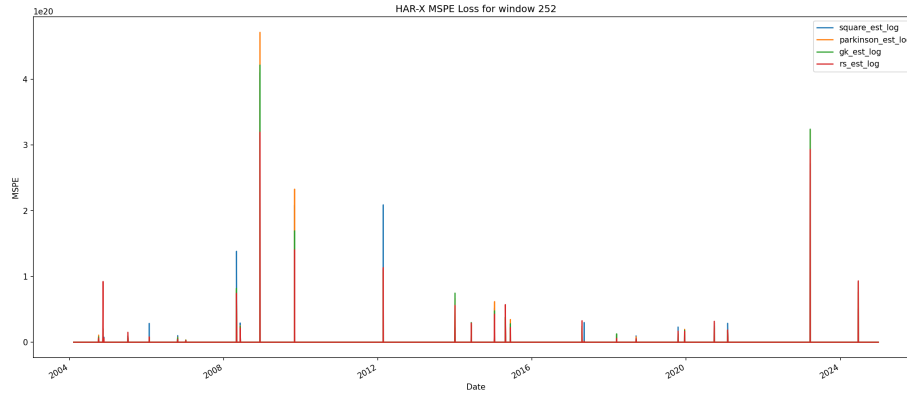


Figure 42: HAR-X MSPE Loss Over Time (Window=252)

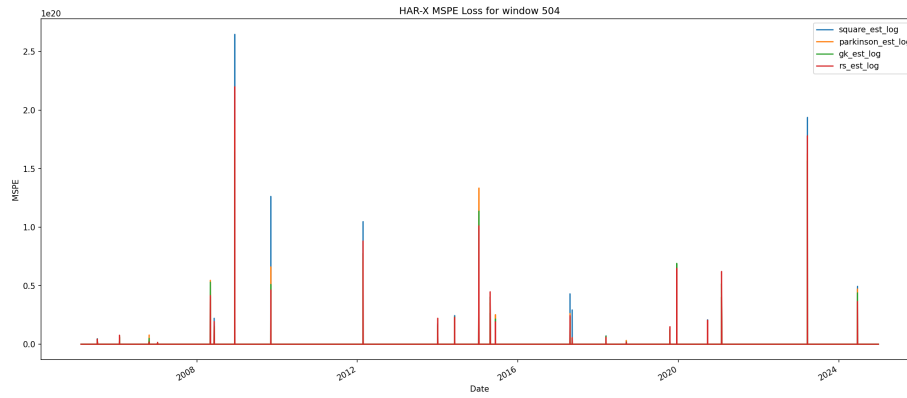


Figure 43: HAR-X MSPE Loss Over Time (Window=504)

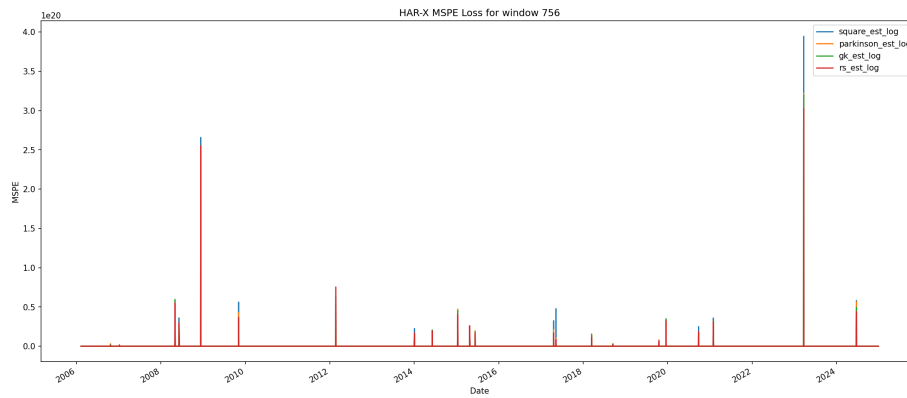


Figure 44: HAR-X MSPE Loss Over Time (Window=756)

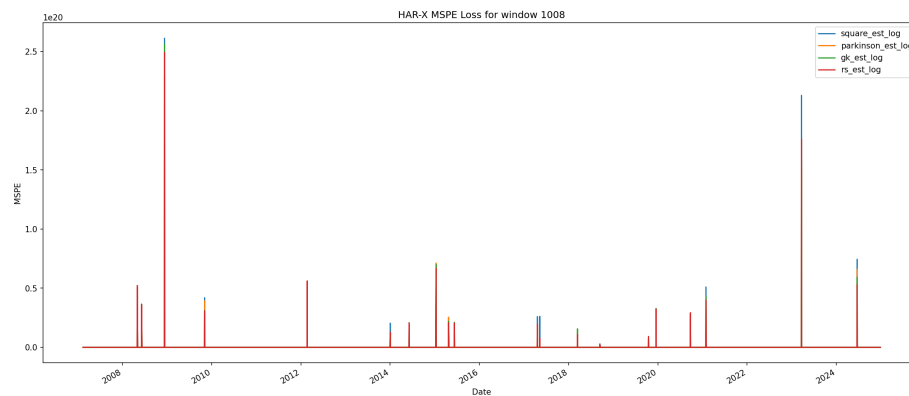


Figure 45: HAR-X MSPE Loss Over Time (Window=1008)

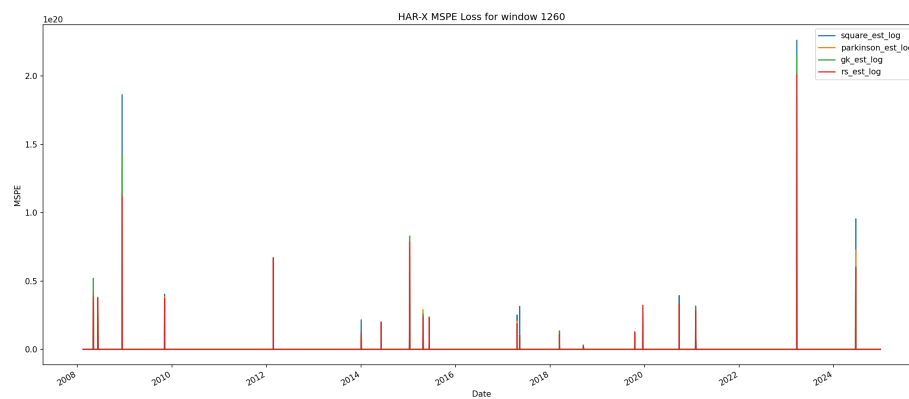


Figure 46: HAR-X MSPE Loss Over Time (Window=1260)

HAR-X Ensemble Model Results

Ensemble Weights

The HAR-X ensemble model combines predictions from multiple estimators using inverse QLIKE weighting. Below are the weights assigned to each estimator for different window sizes.

Table 10: HAR-X Ensemble Model Weights by Window

Window	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
252	0.25	0.25	0.25	0.25
504	0.25	0.25	0.25	0.25
756	0.333	0.333	0.333	0
1008	0.25	0.25	0.25	0.25
1260	0.25	0.25	0.25	0.25

HAR-X Ensemble Performance Summary

Table 11: HAR-X Ensemble Model Performance Metrics

	QLIKE_mean	QLIKE_std	MSPE_mean	MSPE_std
(252, 0)	-1.225	8.825	2.57162e+17	7.02852e+18
(504, 0)	-1.302	7.329	1.82438e+17	4.15148e+18
(756, 0)	-1.306	6.934	1.55059e+17	3.73009e+18
(1008, 0)	-1.213	6.873	1.77289e+17	4.4507e+18
(1260, 0)	-1.207	7.003	1.62539e+17	3.18588e+18

Table 12: HAR-X Ensemble Model Ljung-Box Test Results (Part 1/2)

	lb_stat_10	lb_p_10	lb_stat_20	lb_p_20
(252, 0)	8.159	0.613	13.879	0.837
(252, 1)	8.159	0.613	13.879	0.837
(504, 0)	5.772	0.834	11.948	0.918
(504, 1)	5.772	0.834	11.948	0.918
(756, 0)	9.522	0.483	14.221	0.819
(756, 1)	9.522	0.483	14.221	0.819
(1008, 0)	12.137	0.276	16.168	0.706
(1008, 1)	12.137	0.276	16.168	0.706
(1260, 0)	10.735	0.379	15.059	0.773
(1260, 1)	10.735	0.379	15.059	0.773

Table 12: HAR-X Ensemble Model Ljung-Box Test Results (Part 2/2)

	white_noise_flag	lb_lags_used	n_obs	name
(252, 0)	True	10	4757	
(252, 1)	True	20	4757	
(504, 0)	True	10	4505	
(504, 1)	True	20	4505	
(756, 0)	True	10	4253	
(756, 1)	True	20	4253	
(1008, 0)	True	10	4001	
(1008, 1)	True	20	4001	
(1260, 0)	True	10	3749	
(1260, 1)	True	20	3749	

HAR-X Ensemble Predictions and Loss Metrics

HAR-X Ensemble Model: Predictions vs True RV (Window=252)

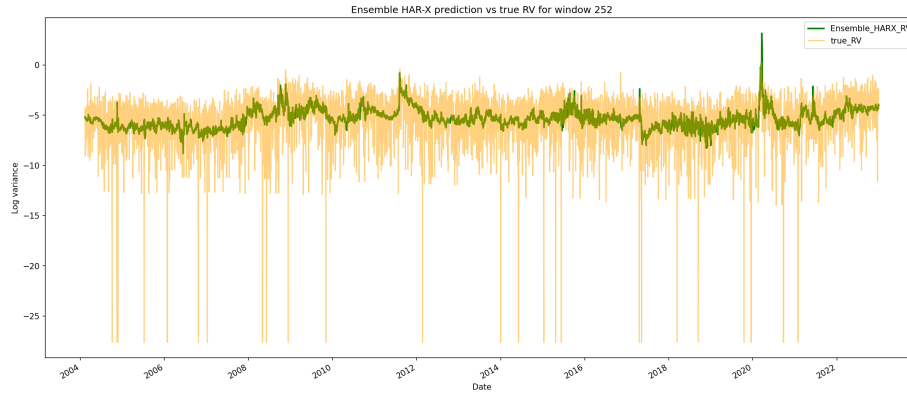


Figure 47: HAR-X Ensemble Model: Predictions vs True RV (Window=252)

HAR-X Ensemble Model: Predictions vs True RV (Window=504)

HAR-X Ensemble Model: Predictions vs True RV (Window=756)

HAR-X Ensemble Model: Predictions vs True RV (Window=1008)

HAR-X Ensemble Model: Predictions vs True RV (Window=1260)

HAR-X Ensemble QLIKE Loss (Window=252)

HAR-X Ensemble QLIKE Loss (Window=504)

HAR-X Ensemble QLIKE Loss (Window=756)

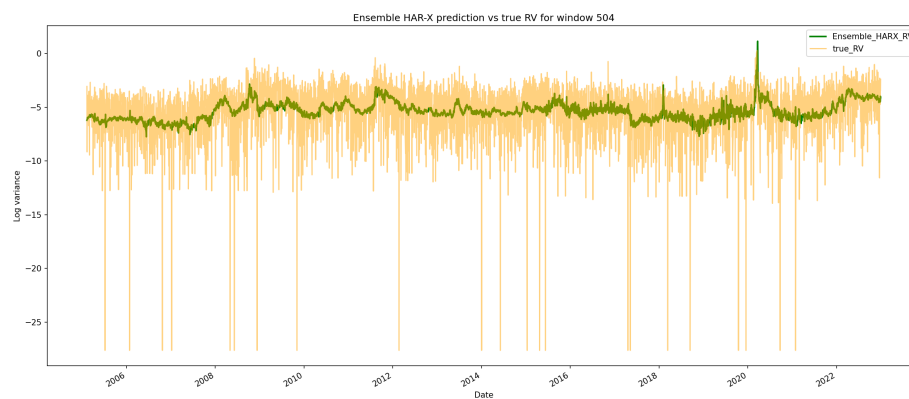


Figure 48: HAR-X Ensemble Model: Predictions vs True RV (Window=504)

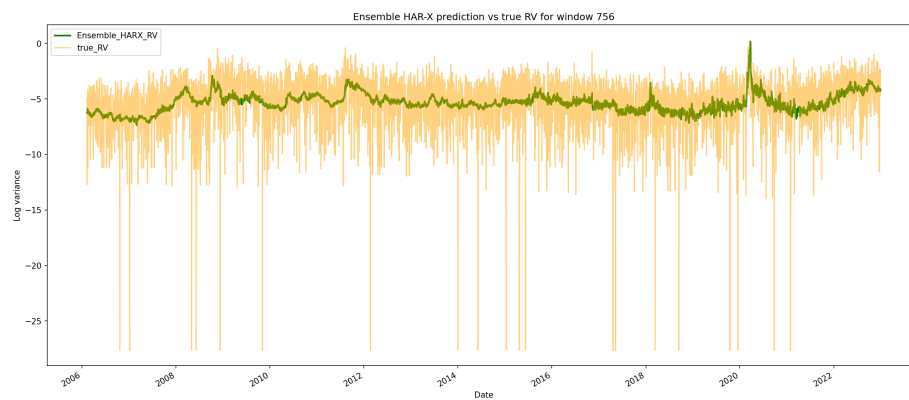


Figure 49: HAR-X Ensemble Model: Predictions vs True RV (Window=756)

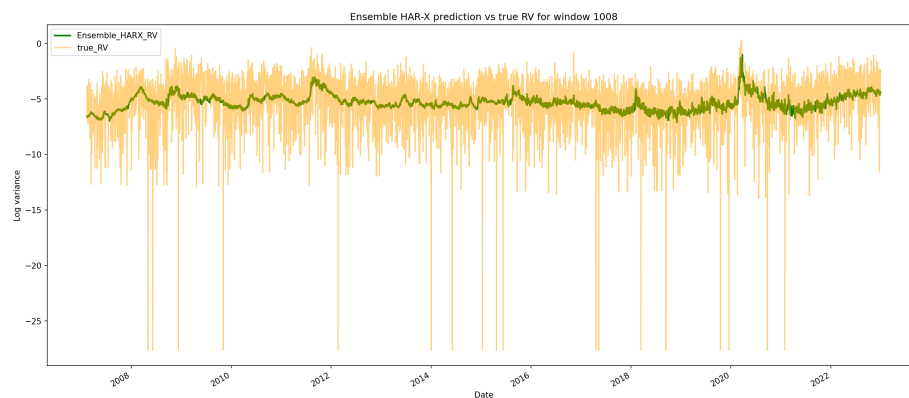


Figure 50: HAR-X Ensemble Model: Predictions vs True RV (Window=1008)

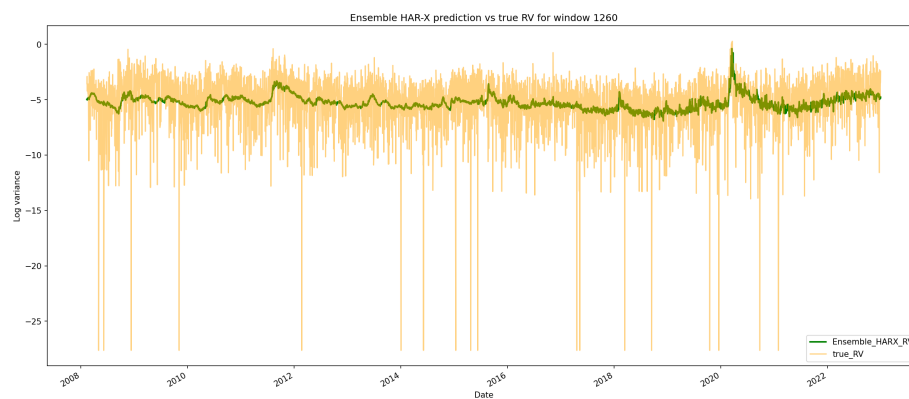


Figure 51: HAR-X Ensemble Model: Predictions vs True RV (Window=1260)

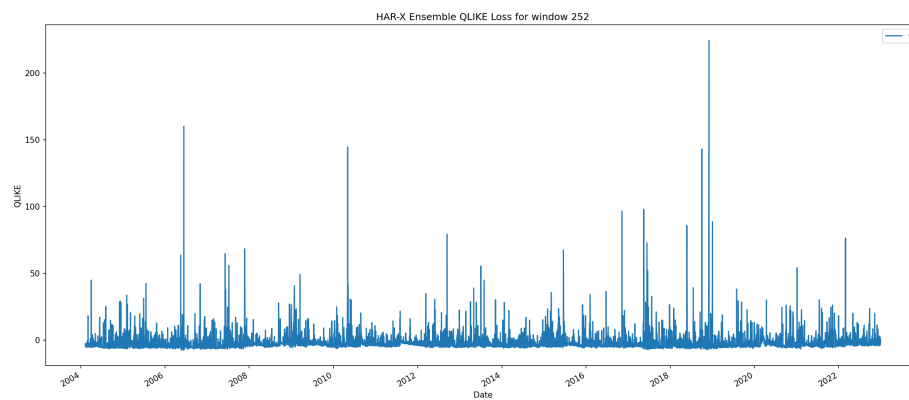


Figure 52: HAR-X Ensemble QLIKE Loss (Window=252)

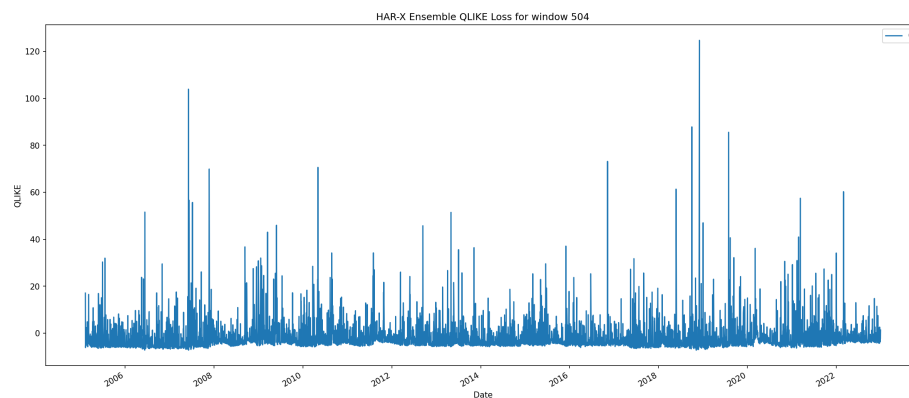


Figure 53: HAR-X Ensemble QLIKE Loss (Window=504)

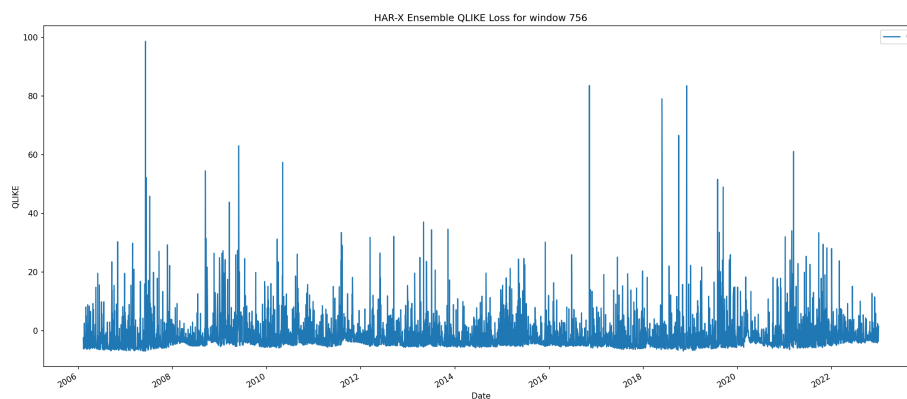


Figure 54: HAR-X Ensemble QLIKE Loss (Window=756)

HAR-X Ensemble QLIKE Loss (Window=1008)

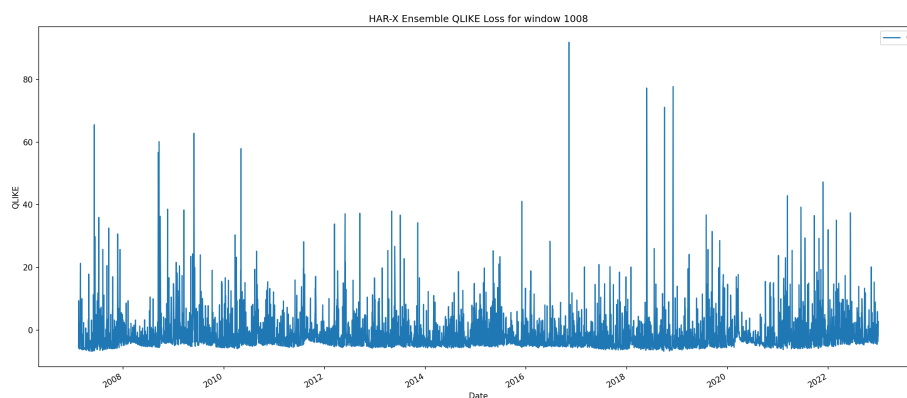


Figure 55: HAR-X Ensemble QLIKE Loss (Window=1008)

HAR-X Ensemble QLIKE Loss (Window=1260)

HAR-X Ensemble MSPE Loss (Window=252)

HAR-X Ensemble MSPE Loss (Window=504)

HAR-X Ensemble MSPE Loss (Window=756)

HAR-X Ensemble MSPE Loss (Window=1008)

HAR-X Ensemble MSPE Loss (Window=1260)

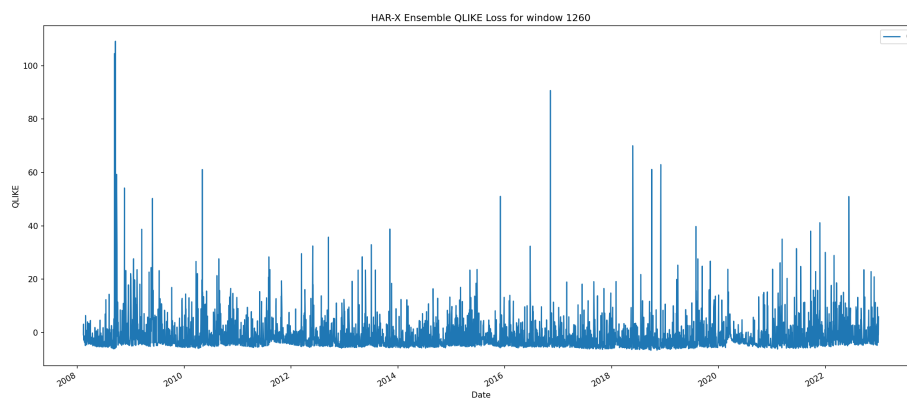


Figure 56: HAR-X Ensemble QLIKE Loss (Window=1260)

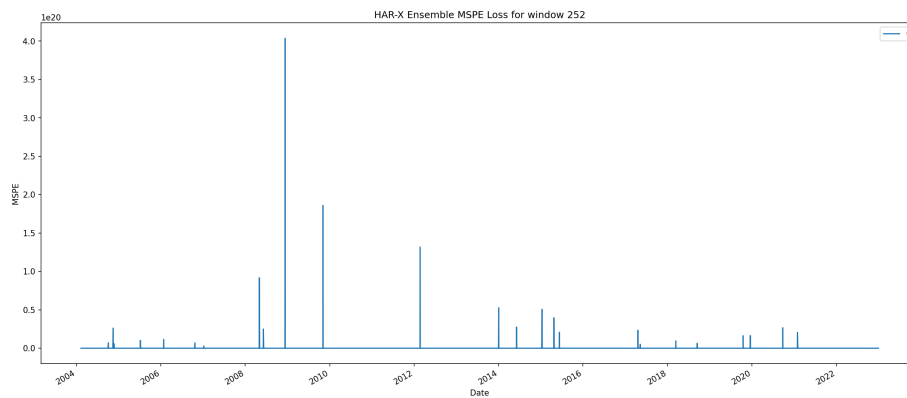


Figure 57: HAR-X Ensemble MSPE Loss (Window=252)

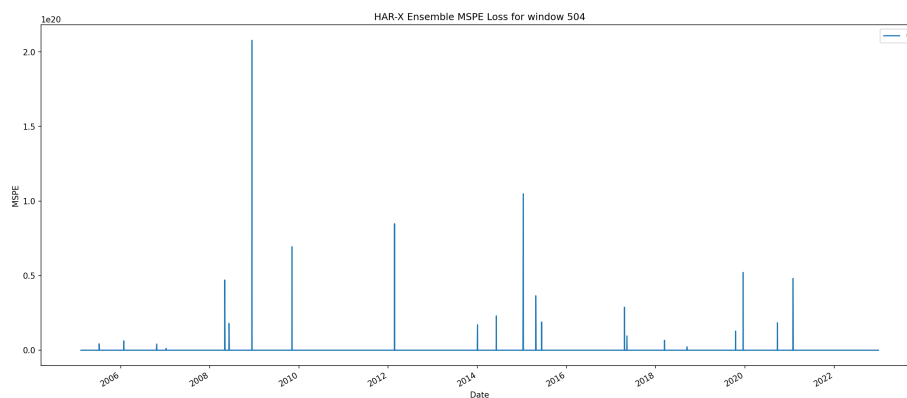


Figure 58: HAR-X Ensemble MSPE Loss (Window=504)

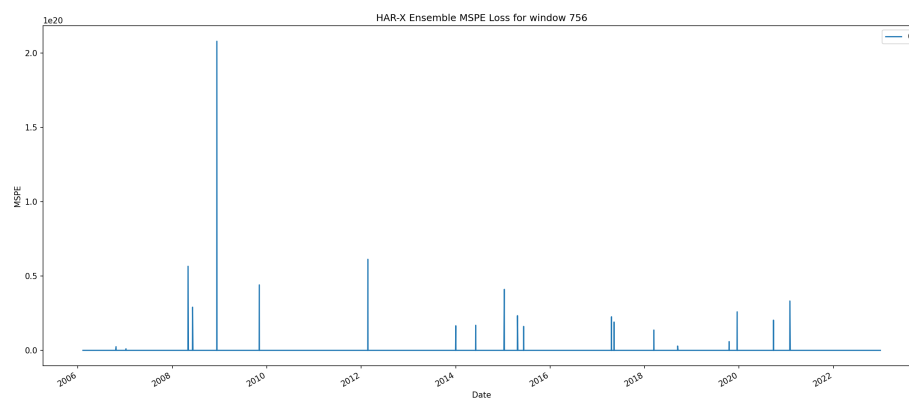


Figure 59: HAR-X Ensemble MSPE Loss (Window=756)

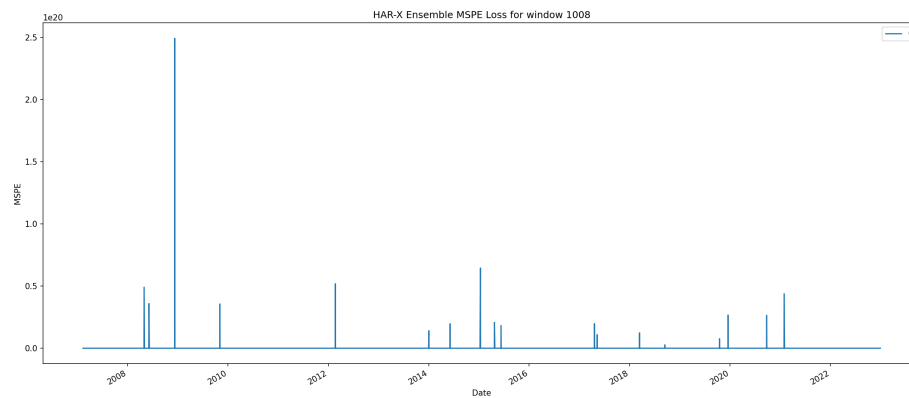


Figure 60: HAR-X Ensemble MSPE Loss (Window=1008)

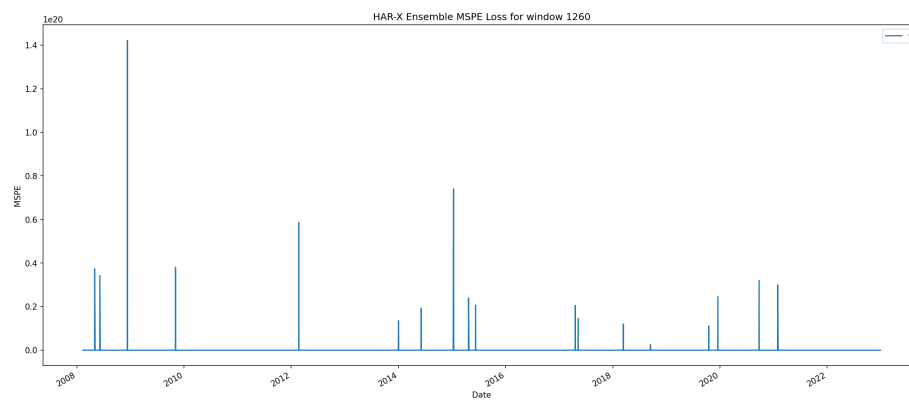


Figure 61: HAR-X Ensemble MSPE Loss (Window=1260)

HAR-X Diebold-Mariano Test

The Diebold-Mariano (DM) test evaluates whether there is a statistically significant difference in predictive accuracy between two HAR-X models. Here we compare Windows 504 and 756.

HAR-X DM Test Results: Window 504 vs Window 756

Metric	Value
DM Statistic	nan
P-value	nan
Better Model	None (No significant difference)
Significant?	0.0000
Alpha	0.0500
Observations	4753.0000

Interpretation: With p-value = nan, we fail to reject the null hypothesis of equal predictive accuracy. This indicates no statistically significant difference between Window 504 and 756 for the HAR-X model.

Report generation completed

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Machine Learning Models

This section evaluates five state-of-the-art machine learning algorithms for volatility forecasting:

- **Random Forest (RF)**: Ensemble of decision trees
- **Gradient Boosting Machine (GBM)**: Sequential tree learning with error correction
- **XGBoost**: Optimized gradient boosting with GPU support
- **LightGBM**: Fast gradient boosting with leaf-wise growth
- **CatBoost**: Boosting optimized for categorical features

Each model is trained on a rolling window of historical volatility estimators and exogenous variables, then evaluated on out-of-sample predictions across three window sizes (252, 504, 756 days).

RF Model Performance

Random Forest (RF) is an ensemble method that builds multiple decision trees and averages their predictions. It's robust to outliers and naturally captures non-linear relationships in volatility forecasting without requiring feature scaling.

Table: RF Performance Summary Across Windows (Part 1/2)

	Window	QLIKE	QLIKE Std	MSPE
0	252	-3.786	1.977	0.165
1	756	-3.904	2.285	0.009

Table: RF Performance Summary Across Windows (Part 2/2)

	MSPE Std	RMSE	Dir. Accuracy
0	2.312	0.006	99.80%
1	0.103	0.001	99.78%

RF Predictions vs True RV The following plots compare predicted volatility against true realized volatility for each rolling window.

RF: Predictions vs True RV (Window=252)

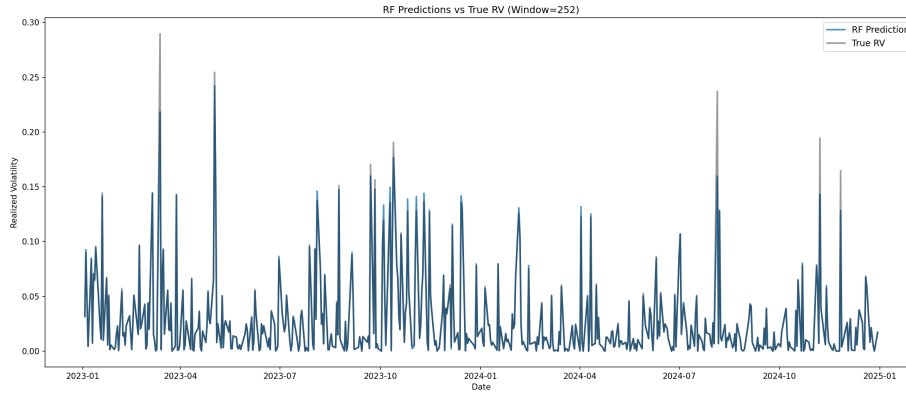


Figure 62: RF: Predictions vs True RV (Window=252)

RF: Predictions vs True RV (Window=756)

RF Loss Metrics Over Time QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude for the ML model across different training periods.

RF QLIKE Loss Over Time (Window=252)

RF QLIKE Loss Over Time (Window=756)

RF MSPE Loss Over Time (Window=252)

RF MSPE Loss Over Time (Window=756)

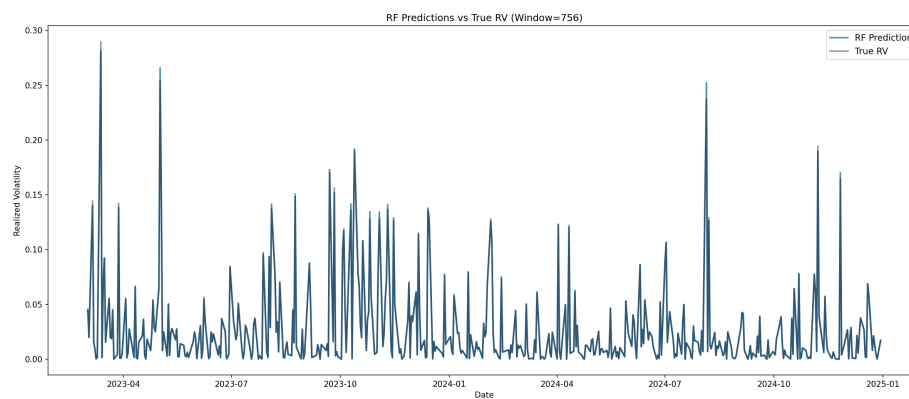


Figure 63: RF: Predictions vs True RV (Window=756)

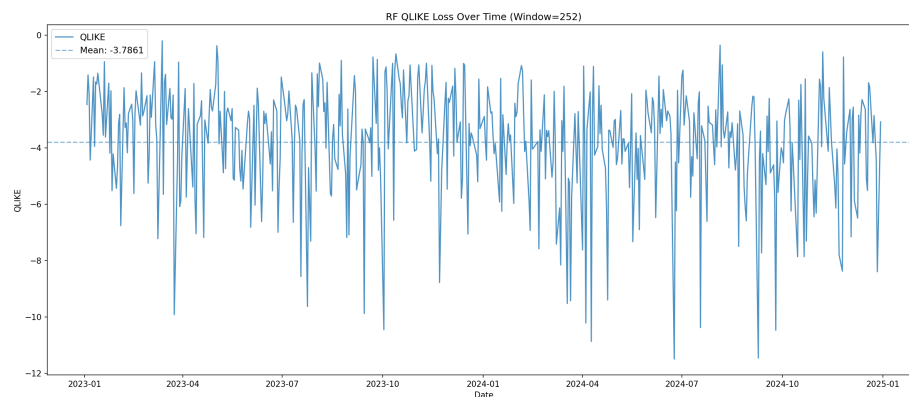


Figure 64: RF QLIKE Loss Over Time (Window=252)

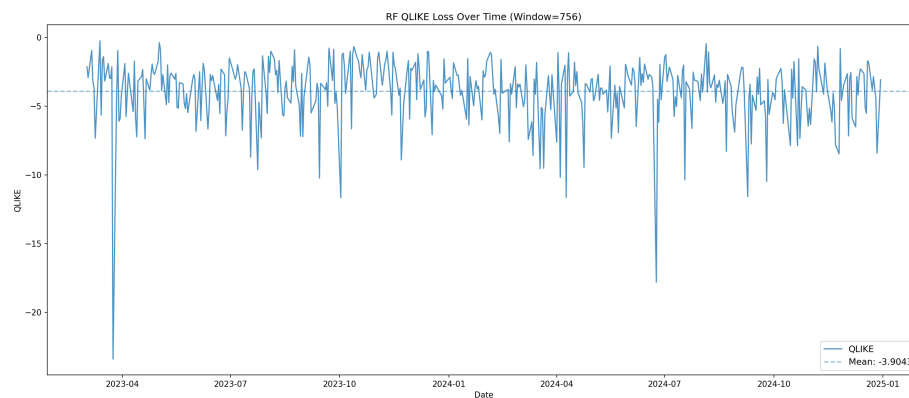


Figure 65: RF QLIKE Loss Over Time (Window=756)

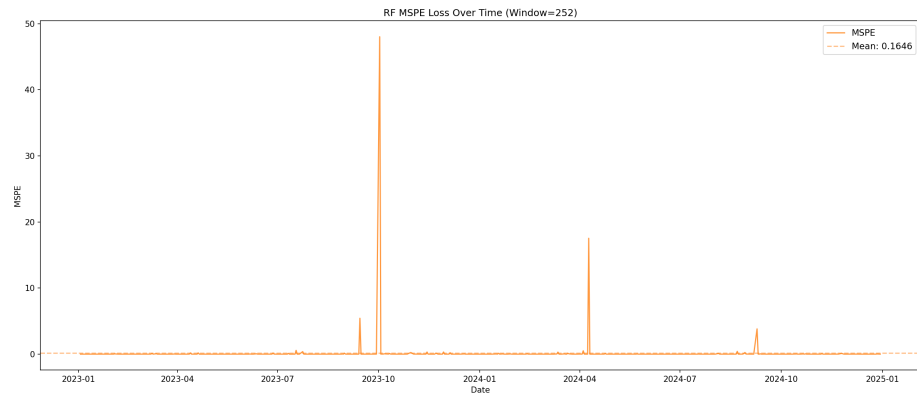


Figure 66: RF MSPE Loss Over Time (Window=252)

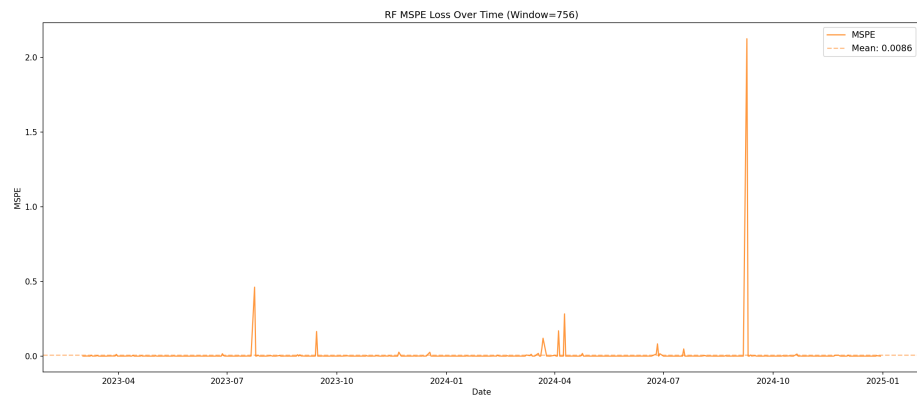


Figure 67: RF MSPE Loss Over Time (Window=756)

GBM Model Performance

Gradient Boosting Machine (GBM) sequentially builds decision trees, with each tree correcting the errors of previous ones. This incremental approach often provides superior performance for forecasting tasks by explicitly learning residual patterns.

Table: GBM Performance Summary Across Windows (Part 1/2)

	Window	QLIKE	QLIKE Std	MSPE
0	252	-3.749	2.046	0.035
1	756	-3.914	2.368	0.012

Table: GBM Performance Summary Across Windows (Part 2/2)

	MSPE Std	RMSE	Dir. Accuracy
0	0.275	0.004	99.60%
1	0.164	0.002	100.00%

GBM Predictions vs True RV The following plots compare predicted volatility against true realized volatility for each rolling window.

GBM: Predictions vs True RV (Window=252)

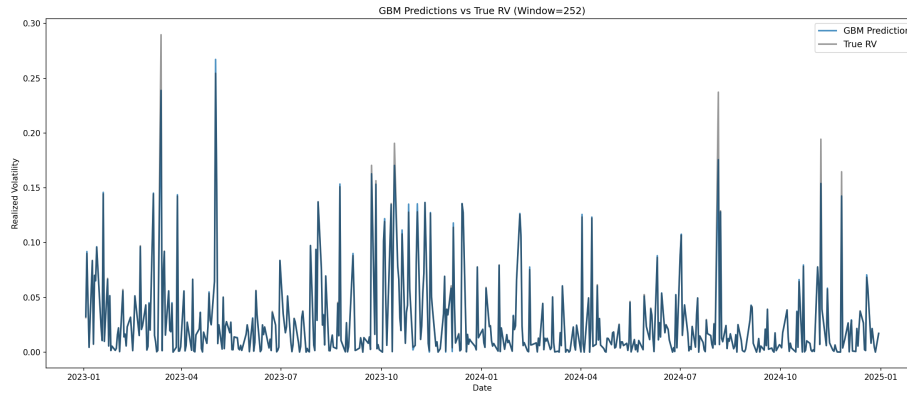


Figure 68: GBM: Predictions vs True RV (Window=252)

GBM: Predictions vs True RV (Window=756)

GBM Loss Metrics Over Time QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window.

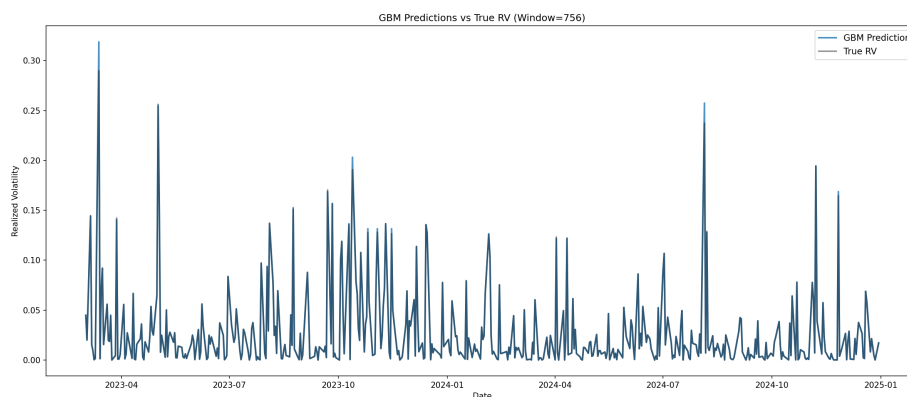


Figure 69: GBM: Predictions vs True RV (Window=756)

These metrics help assess forecast calibration and error magnitude for the ML model across different training periods.

GBM QLIKE Loss Over Time (Window=252)

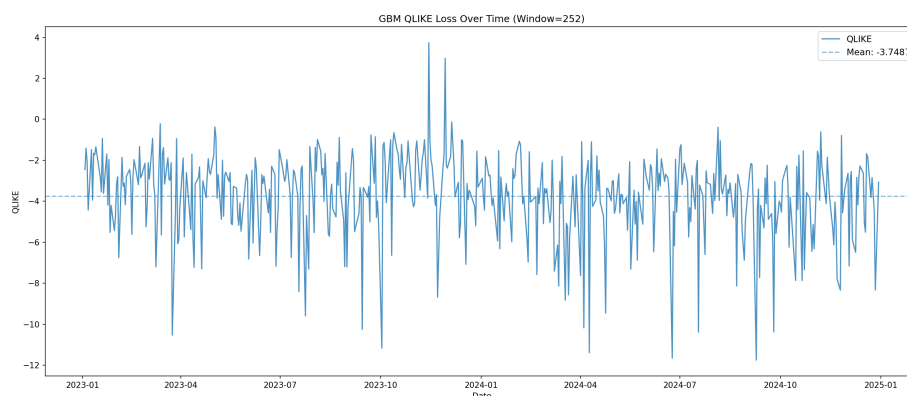


Figure 70: GBM QLIKE Loss Over Time (Window=252)

GBM QLIKE Loss Over Time (Window=756)

GBM MSPE Loss Over Time (Window=252)

GBM MSPE Loss Over Time (Window=756)

XGBOOST Model Performance

XGBoost (eXtreme Gradient Boosting) is an optimized gradient boosting implementation with GPU acceleration support. It includes regularization to pre-

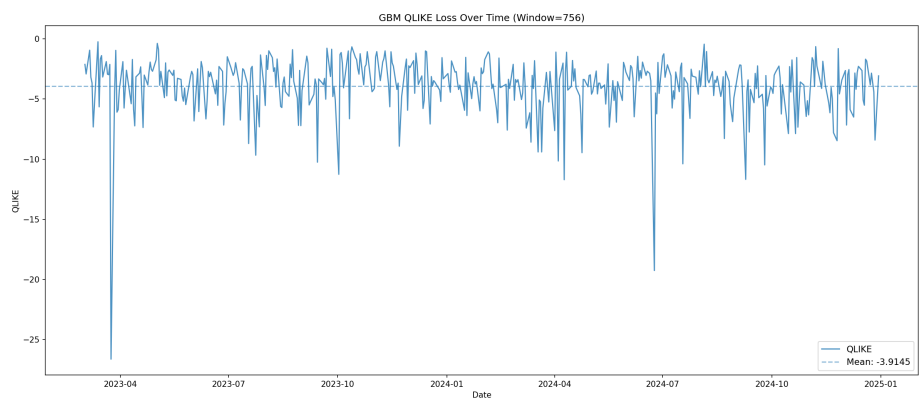


Figure 71: GBM QLIKE Loss Over Time (Window=756)

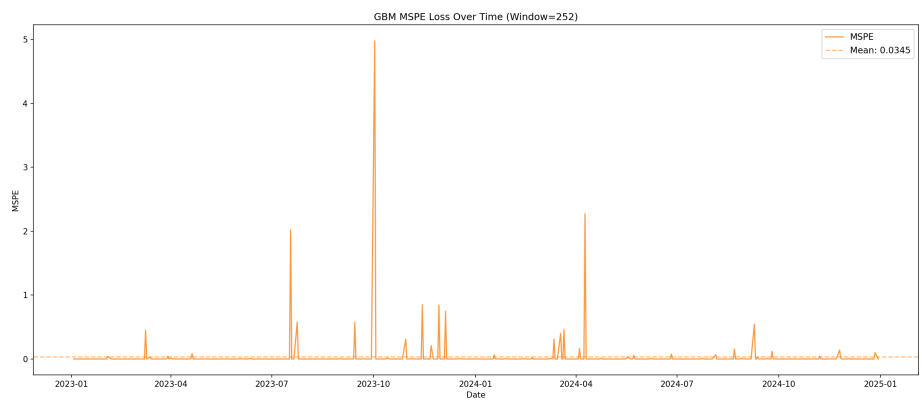


Figure 72: GBM MSPE Loss Over Time (Window=252)

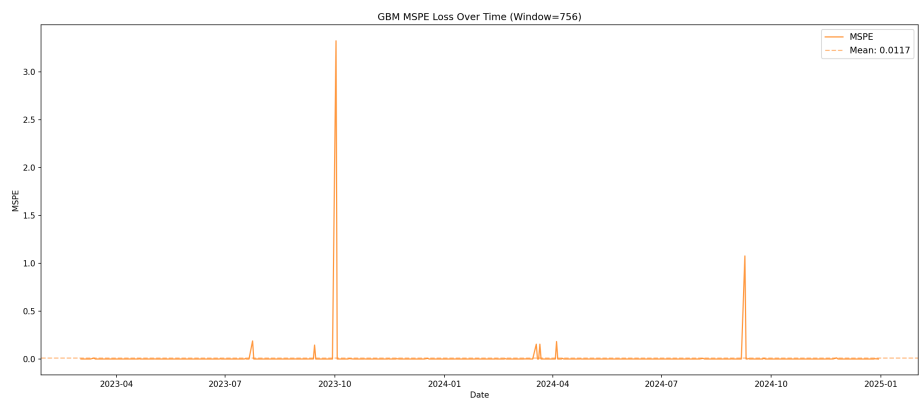


Figure 73: GBM MSPE Loss Over Time (Window=756)

vent overfitting and has proven highly effective for time series forecasting competitions.

Table: XGBOOST Performance Summary Across Windows (Part 1/2)

	Window	QLIKE	QLIKE Std	MSPE
0	252	80.832	1331.57	0.104
1	756	-3.681	2.607	0.052

Table: XGBOOST Performance Summary Across Windows (Part 2/2)

	MSPE Std	RMSE	Dir. Accuracy
0	0.659	0.014	97.38%
1	0.205	0.01	98.03%

XGBOOST Predictions vs True RV The following plots compare predicted volatility against true realized volatility for each rolling window.

XGBOOST: Predictions vs True RV (Window=252)

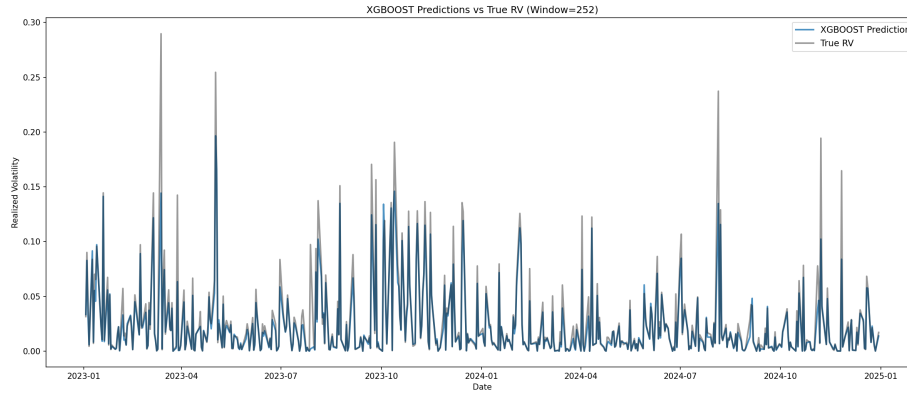


Figure 74: XGBOOST: Predictions vs True RV (Window=252)

XGBOOST: Predictions vs True RV (Window=756)

XGBOOST Loss Metrics Over Time QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude for the ML model across different training periods.

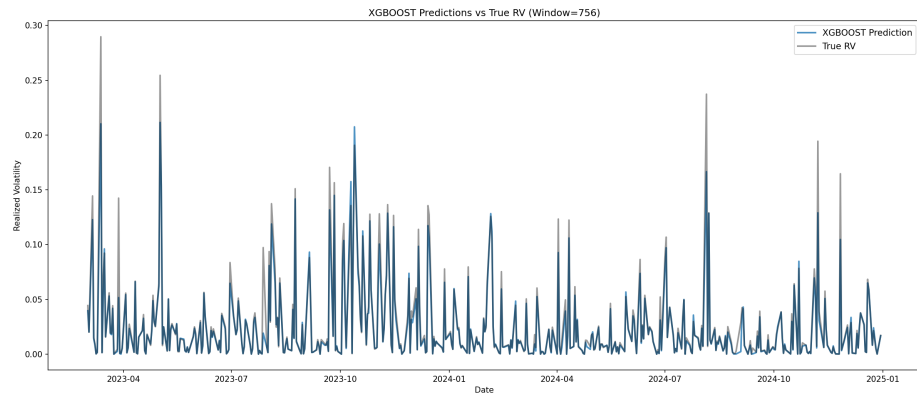


Figure 75: XGBOOST: Predictions vs True RV (Window=756)

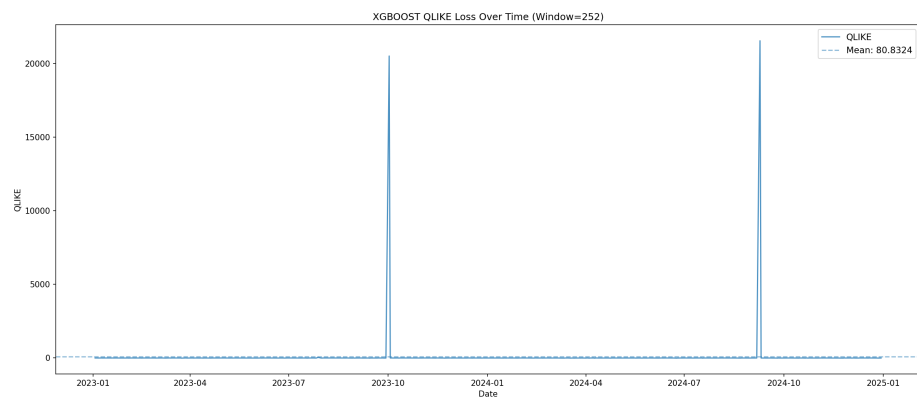


Figure 76: XGBOOST QLIKE Loss Over Time (Window=252)

XGBOOST QLIKE Loss Over Time (Window=252)

XGBOOST QLIKE Loss Over Time (Window=756)

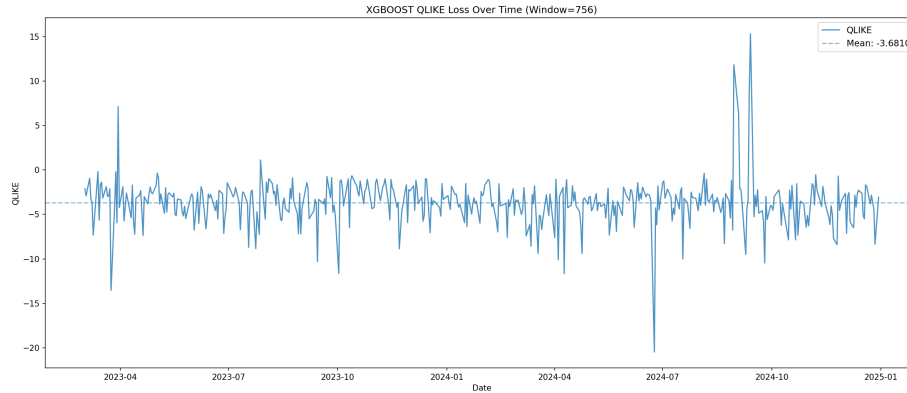


Figure 77: XGBOOST QLIKE Loss Over Time (Window=756)

XGBOOST MSPE Loss Over Time (Window=252)

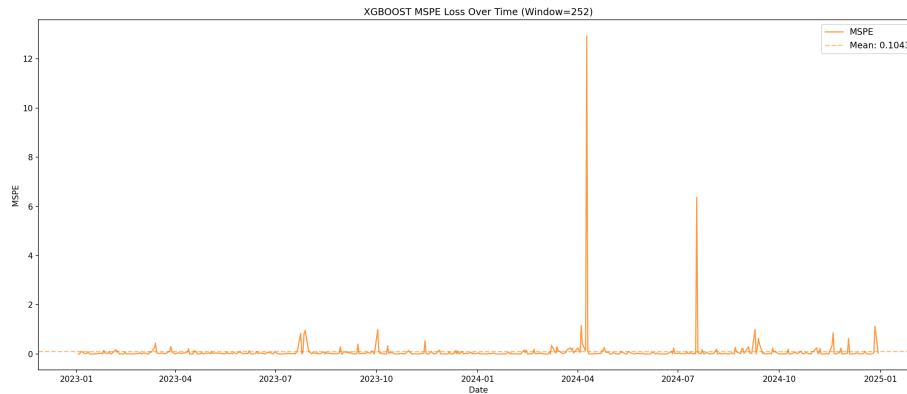


Figure 78: XGBOOST MSPE Loss Over Time (Window=252)

XGBOOST MSPE Loss Over Time (Window=756)

LIGHTGBM Model Performance

LightGBM (Light Gradient Boosting Machine) is a fast, distributed gradient boosting framework with GPU support. It uses leaf-wise tree growth for better accuracy and is especially efficient with large feature sets.

Table: LIGHTGBM Performance Summary Across Windows (Part 1/2)

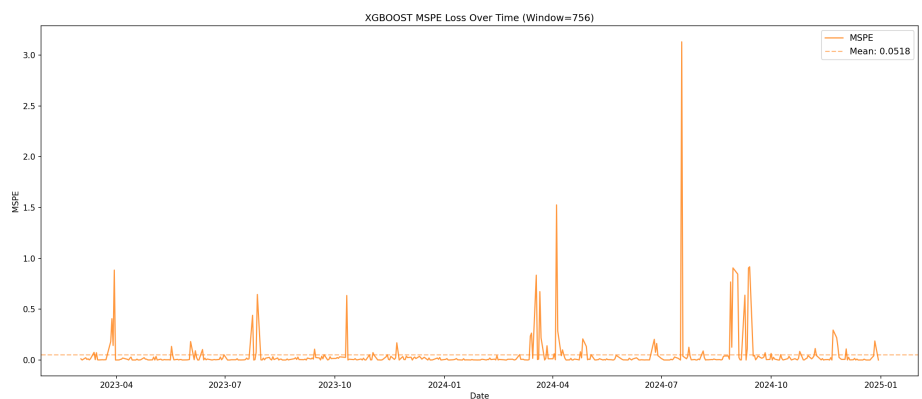


Figure 79: XGBOOST MSPE Loss Over Time (Window=756)

	Window	QLIKE	QLIKE Std	MSPE
0	252	-3.603	1.881	3.058
1	756	-3.428	8.219	0.063

Table: LIGHTGBM Performance Summary Across Windows (Part 2/2)

	MSPE Std	RMSE	Dir. Accuracy
0	39.313	0.014	97.18%
1	0.405	0.009	98.25%

LIGHTGBM Predictions vs True RV The following plots compare predicted volatility against true realized volatility for each rolling window.

LIGHTGBM: Predictions vs True RV (Window=252)

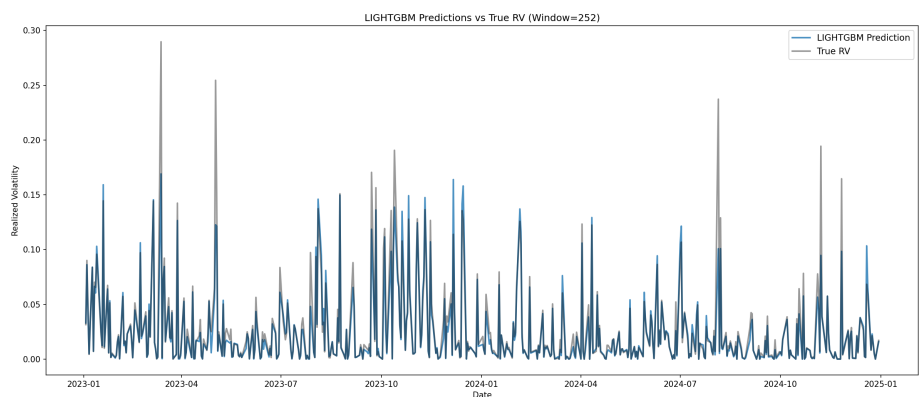


Figure 80: LIGHTGBM: Predictions vs True RV (Window=252)

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LIGHTGBM: Predictions vs True RV (Window=756)

LIGHTGBM Loss Metrics Over Time QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude for the ML model across different training periods.

LIGHTGBM QLIKE Loss Over Time (Window=252)

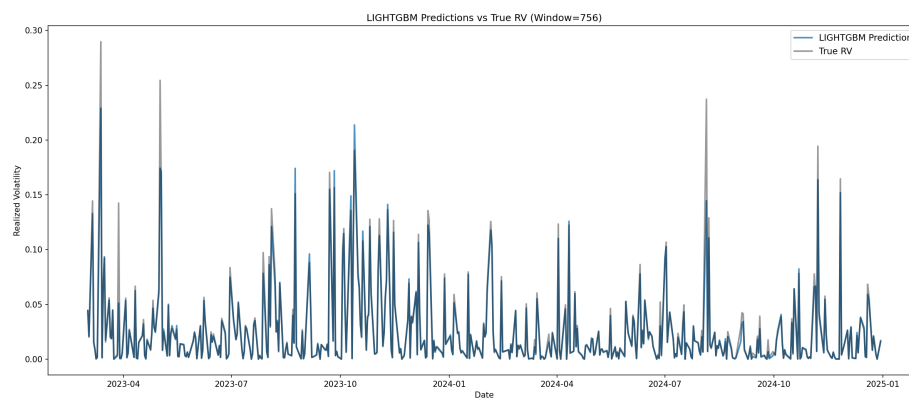


Figure 81: LIGHTGBM: Predictions vs True RV (Window=756)

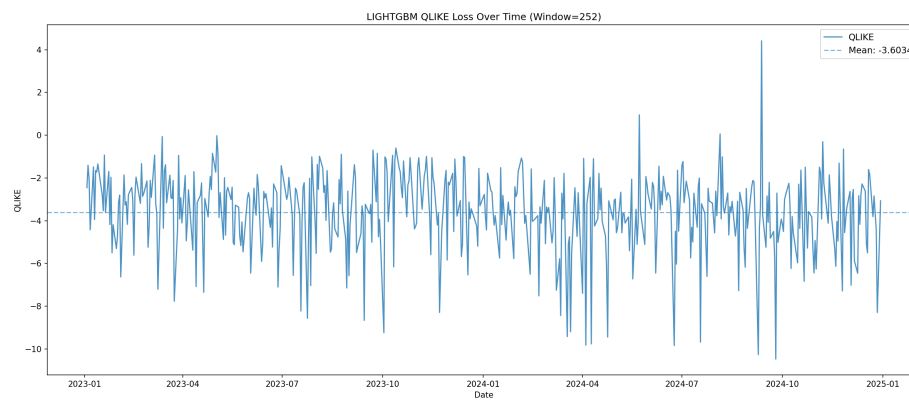


Figure 82: LIGHTGBM QLIKE Loss Over Time (Window=252)

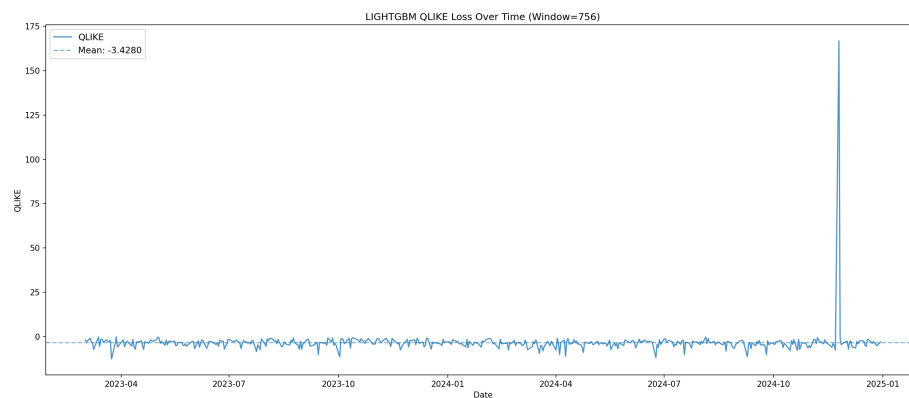


Figure 83: LIGHTGBM QLIKE Loss Over Time (Window=756)

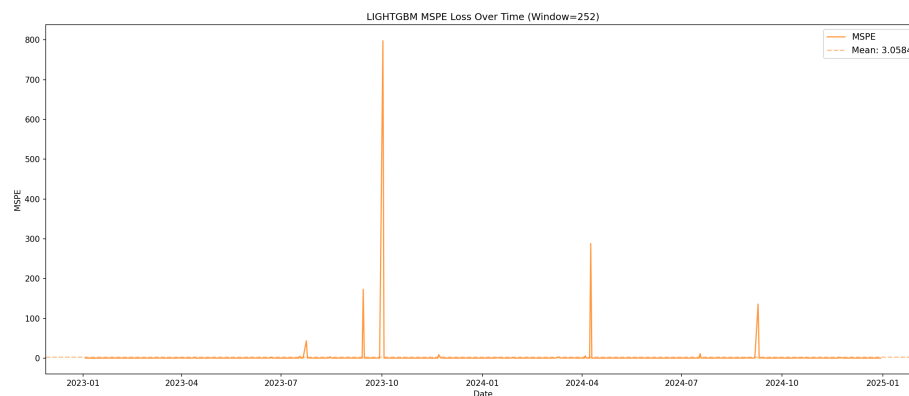


Figure 84: LIGHTGBM MSPE Loss Over Time (Window=252)

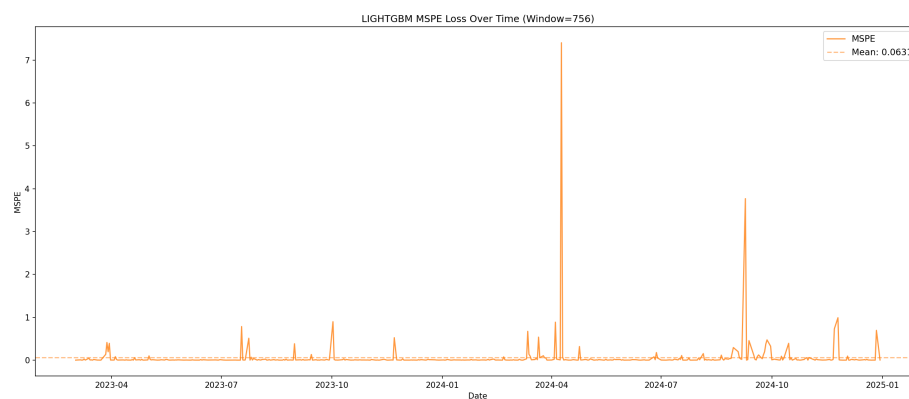


Figure 85: LIGHTGBM MSPE Loss Over Time (Window=756)

CATBOOST Model Performance

CatBoost (Categorical Boosting) is designed to handle categorical features effectively with GPU acceleration. It uses ordered boosting and symmetric trees to reduce overfitting, making it ideal for heterogeneous financial data.

Table: CATBOOST Performance Summary Across Windows (Part 1/2)

	Window	QLIKE	QLIKE Std	MSPE
0	252	-3.73	1.931	2.269
1	756	-3.85	2.033	0.065

Table: CATBOOST Performance Summary Across Windows (Part 2/2)

	MSPE Std	RMSE	Dir. Accuracy
0	27.11	0.02	98.59%
1	0.605	0.009	99.34%

CATBOOST Predictions vs True RV The following plots compare predicted volatility against true realized volatility for each rolling window.

CATBOOST: Predictions vs True RV (Window=252)

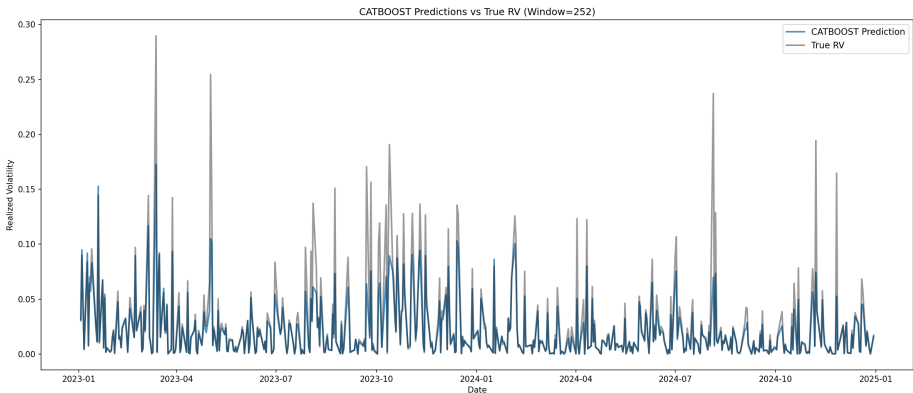


Figure 86: CATBOOST: Predictions vs True RV (Window=252)

CATBOOST: Predictions vs True RV (Window=756)

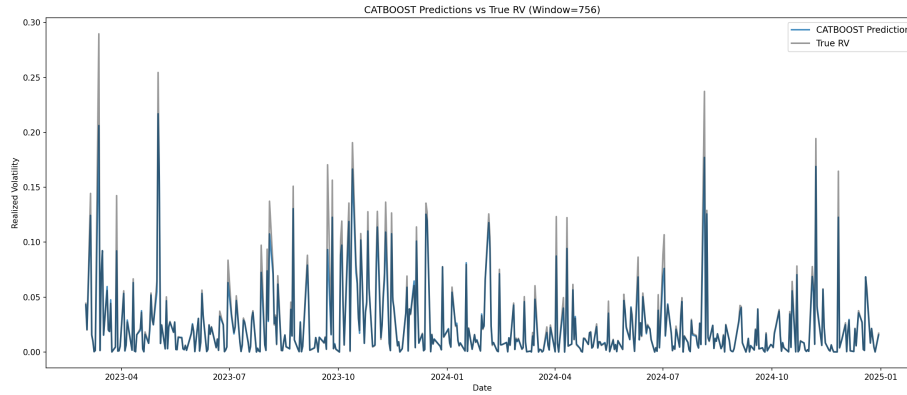


Figure 87: CATBOOST: Predictions vs True RV (Window=756)

CATBOOST Loss Metrics Over Time QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude for the ML model across different training periods.

CATBOOST QLIKE Loss Over Time (Window=252)

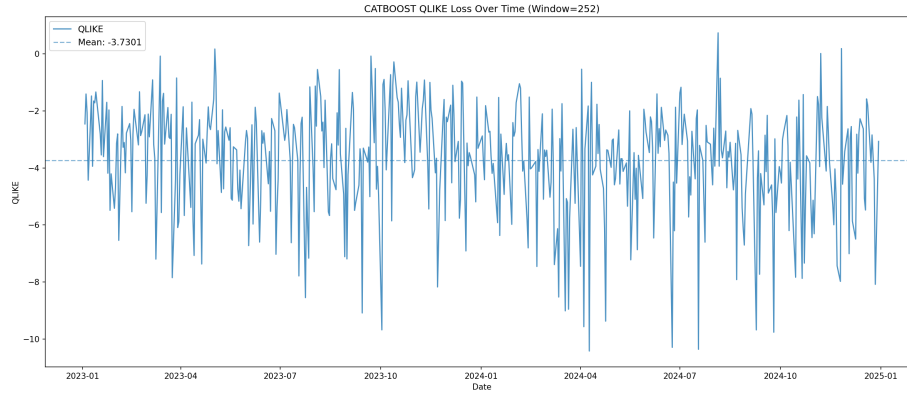


Figure 88: CATBOOST QLIKE Loss Over Time (Window=252)

CATBOOST QLIKE Loss Over Time (Window=756)

CATBOOST MSPE Loss Over Time (Window=252)

CATBOOST MSPE Loss Over Time (Window=756)

ML Models Analysis Completed Successfully

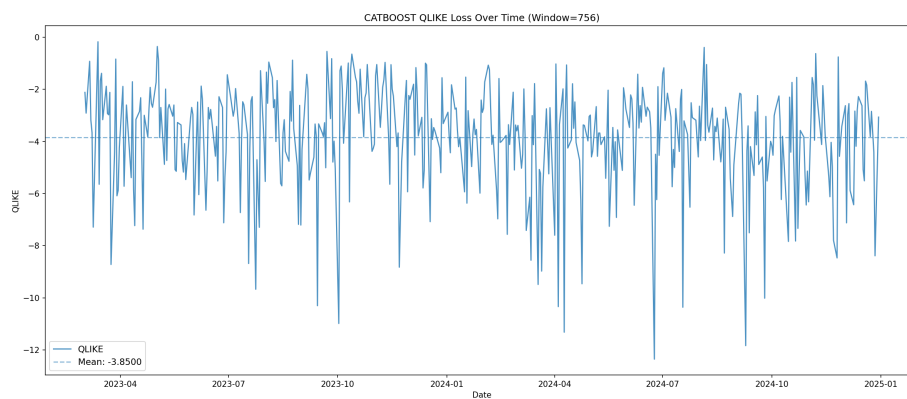


Figure 89: CATBOOST QLIKE Loss Over Time (Window=756)

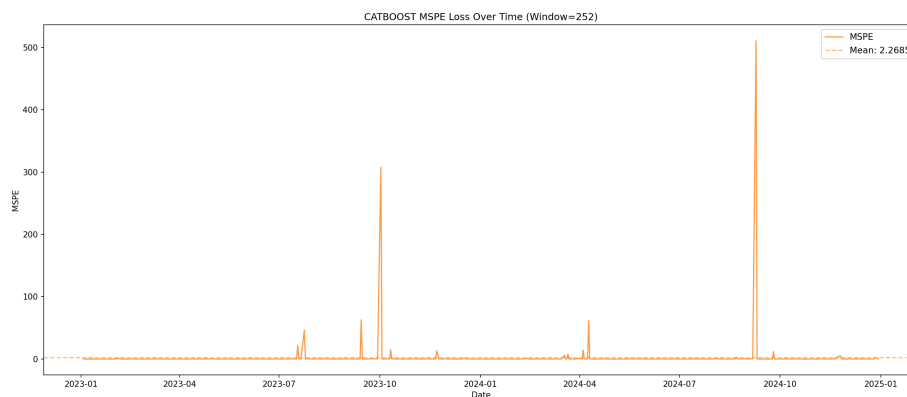


Figure 90: CATBOOST MSPE Loss Over Time (Window=252)

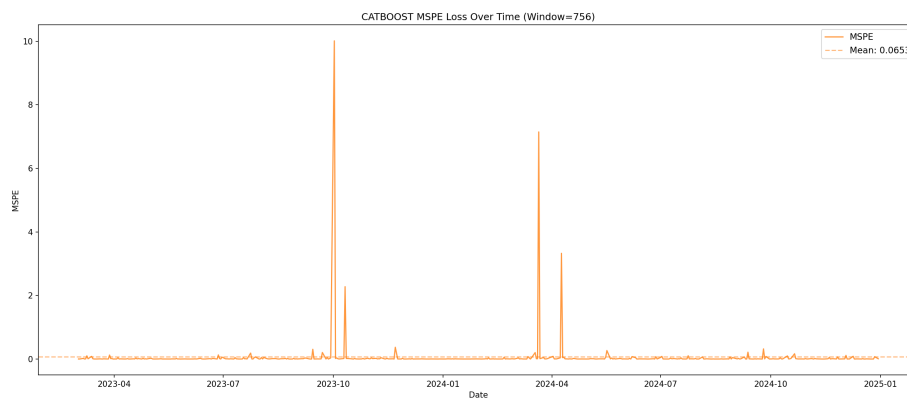


Figure 91: CATBOOST MSPE Loss Over Time (Window=756)

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Temporal Fusion Transformer (TFT) Results

The Temporal Fusion Transformer is a state-of-the-art deep learning architecture for multi-horizon time series forecasting. It combines:

- **Multi-head attention mechanism:** Captures complex temporal dependencies
- **Variable selection networks:** Automatic feature importance learning
- **Gated residual networks:** Non-linear processing with skip connections
- **Quantile forecasting:** Provides prediction intervals (10th, 50th, 90th percentiles)

Key Improvements Implemented: - **Multiple quantiles:** Generates prediction intervals for uncertainty quantification - **Increased model capacity:** Larger hidden sizes and attention heads for better learning - **Enhanced regularization:** Higher dropout to prevent overfitting - **Extended lookback:** 90-day encoder length for capturing longer-term patterns

TFT Architecture Details: - Hidden size: 64 (optimized for dataset size) - Attention heads: 4 - Encoder length: 90 days (quarterly lookback) - Dropout: 0.1 (enhanced regularization) - Quantiles: 0.05, 0.1, 0.25, 0.5, 0.75, 0.9, 0.95 (7 quantiles) - Early stopping: Patience of 20 epochs

TFT Model Performance (Validation Set)

Metric	Value
QLIKE Mean	-2.498336
QLIKE Std	0.911136
QLIKE Min	-3.671221
QLIKE Max	3.131789
MSPE Mean	1846488.368995
MSPE Std	26482412.816841
MSPE Min	0.000001
MSPE Max	564243955.619852
RMSE	0.052859
Directional Accuracy	0.4044 (40.44%)
Training Samples	4523
Validation Samples	530

Figure: TFT Main Performance Analysis (Validation Set)

Figure: TFT QLIKE Distribution Analysis

Figure: TFT MSPE Distribution Analysis

Figure: TFT RMSE Analysis

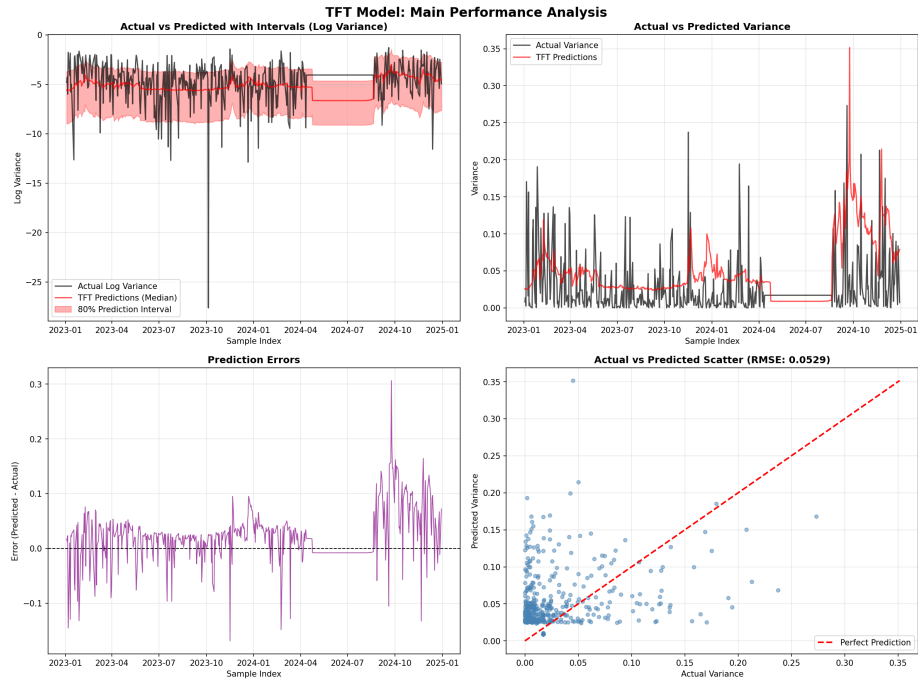


Figure 92: Figure: TFT Main Performance Analysis (Validation Set)

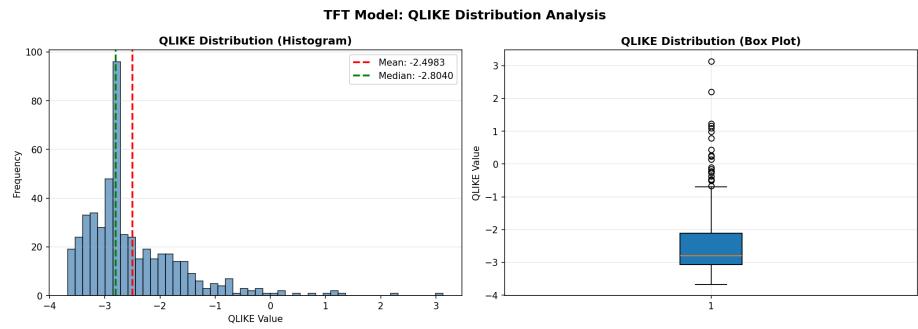


Figure 93: Figure: TFT QLIKE Distribution Analysis

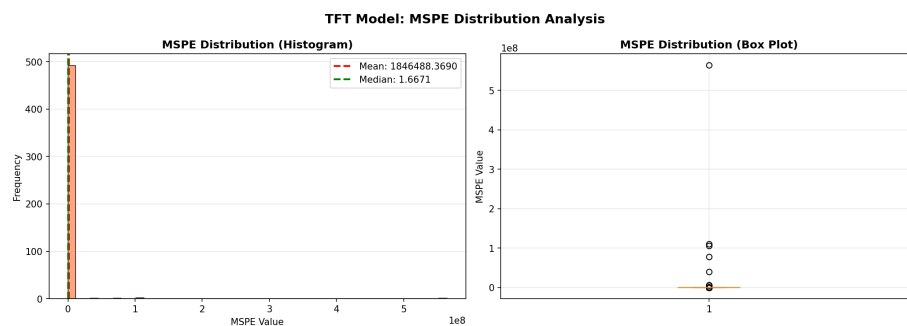


Figure 94: Figure: TFT MSPE Distribution Analysis

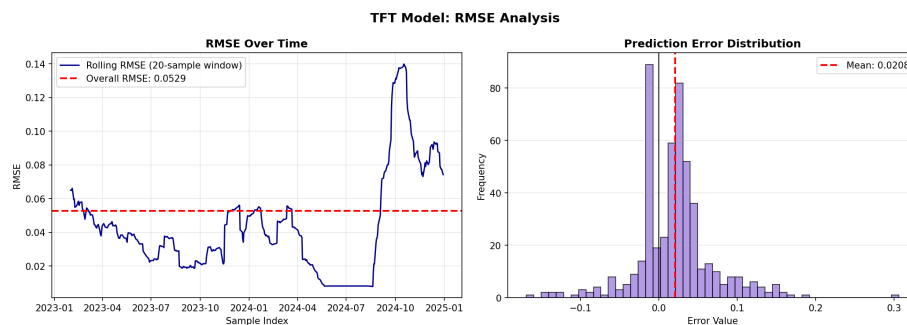


Figure 95: Figure: TFT RMSE Analysis

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